

**A GUIDE TO
THE MANAGEMENT
OF
COMMON MEDICAL
EMERGENCIES
IN ADULTS**

Edited by

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Division of Emergency Medicine
University of the Witwatersrand

**ELEVENTH EDITION
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A GUIDE TO THE MANAGEMENT OF COMMON MEDICAL EMERGENCIES IN ADULTS

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PREFACE TO ELEVENTH EDITION The editor would like to thank all those who contributed valuable
and useful suggestions towards this 11th edition. A number of chapters have been revised and updated, and
several new sections have been added. Additional suggestions and comments are welcomed, and should be
sent to

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It is our hope that this booklet will serve as a useful guide for the initial management of the emergency
situation

DISCLAIMER

The Authors and Editor have exerted every effort to ensure that the clinical procedures and recommendations described herein are based on current knowledge and state-of-the-art information obtained from acknowledged authorities, texts and journals. However, they cannot be considered absolute and universal recommendations. Each patient situation must be considered individually. The reader is urged to check the package inserts of drugs and equipment and the manufacturer's recommendations for indications, contraindications, proper usage, warnings and precautions before use. The Authors and Editor disclaim responsibility for any adverse effects resulting directly or indirectly from information presented in this booklet, undetected errors or misunderstandings by the readers.

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SECTION 1: CARDIO-RESPIRATORY EMERGENCIES

SECTION 1.1 CARDIAC ARREST

ASSESSMENT

H - "Hazards?" Ensure that the environment is safe, make it safe, or remove patient to where it is safe.

H - "Hello?" Check for responsiveness by tapping and talking to the patient. Check for breathing and pulse. If responsive and/or breathing normally, treat illnesses or injuries as necessary. Get assistance if needed, and reassess patient continuously. **H - "Help!"** If patient is unresponsive and not breathing or having only occasional gasps, call for help and activate the Emergency Medical Services / Resuscitation Team. If no pulse, call for a Defibrillator/AED and start CPR, beginning with chest compressions. If a pulse is present but breathing is ineffective, give rescue breaths (once every 6 seconds). Reassess continuously.

CARDIO-PULMONARY RESUSCITATION (CPR)

No Pulse (or not sure):

Briefly check for the presence or absence of a central pulse in a patient who is unresponsive and not breathing normally. If no definite pulse is felt within 10 seconds, or if uncertain, immediately start chest compressions while awaiting the defibrillator.

Any unresponsive patient who is not breathing or having only occasional gasps requires **immediate activation** of the Emergency Medical Services (EMS) or local Resuscitation Team. Agonal respiration (gaspings) can persist for several minutes after cardiac arrest, and should **not** delay the initiation of resuscitation. Similarly, seizure-like activity may be an early manifestation of cardiac arrest. Know your local EMS/Resuscitation Team telephone numbers! Ensure that healthcare professional assistance with defibrillation and/or advanced life support capabilities are activated, and that the exact location of the patient has been confirmed.

Start Compressions:

Place patient supine on a **firm**, flat surface. Position yourself directly above the patient and place the heel of both hands, one on top of the other, on the centre of the chest on the lower half of the sternum just above the xiphisternum. Keep your elbows straight and your shoulders directly above your hands. If the patient is on a bed or stretcher and cannot safely be moved, it is essential that you **stand on a step or chair** to ensure that you are positioned well above the patient. If the patient is lying on a mattress, place a cardiac arrest board under the patient's chest, ensuring that you are able to kneel on the board alongside the patient's shoulder. Compress sternum to a depth of **at least** 5cm at a rate of 100 - 120 per minute (almost 2 compressions per second). Push hard and push fast, and allow complete chest recoil after each compression. Minimize interruptions.

After 30 chest compressions, open the airway and give 2 rescue breaths. This can be done using a bag-valve-resuscitator, or face mask, or mouth-to-mouth using a protective barrier device. Provide supplemental oxygen if available.

If for whatever reason ventilations cannot be given, perform **continuous uninterrupted chest compressions** until a suitable device becomes available.

Breaths:

To perform effective (chest rising) rescue breaths, open the airway by lifting the bony part of the chin with the fingers of one hand, while placing your other hand on the patient's forehead and tilting the head backwards (**Head tilt - Chin lift Manoeuvre**). This will lift the jaw and tongue off the posterior pharyngeal wall, opening the airway. Do not tilt the head backwards if a neck injury is suspected - instead place your fingers behind the jaw on each side and lift the jaw upwards while opening the mouth with your thumbs (**Jaw - Thrust Manoeuvre**). Quickly look inside the mouth and remove any **vomitus or foreign matter** if present. Remove dentures **only** if they cannot rapidly be managed into place.

If a **face mask** is being used for ventilation, a tight seal around the mouth and nose is **mandatory** while keeping the airway open with the Jaw-Thrust Manoeuvre. If the correct size oropharyngeal and/or nasopharyngeal airway is available, this could be inserted. Ensure that the chest rises with each ventilation.

If only a plastic mouth-to-mouth **protective barrier device** is available, place the device between the patient's teeth. Placing your hands under the plastic, lift the jaw upwards with one hand while keeping the nostrils closed using the other hand which is on the patient's forehead. Form a tight mouth-to-mouth seal over the device, and ensure that the chest rises with each breath given.

If two rescuers are present, one rescuer compresses the chest, while the other rescuer gives 2 breaths after every 30 compressions. Pause for the ventilation unless the patient has an advanced airway in place (tracheal tube, laryngeal tube, laryngeal mask airway or Combitube/Easytube), in which case **uninterrupted compressions** are performed with an interposed ventilation given once every 6 seconds.

Focus on high quality CPR:

- Push **hard** (at least 5cm, but not more than 6cm)
- Push **fast** (at least 100/min, but not more than 120/min)
- Allow **complete** chest recoil after each compression
- **Minimize interruptions** in compressions. Avoid interrupting CPR for more than 10 seconds if possible.
- **Avoid** excessive ventilation
- **Rotate** compressors every 2 minutes

Defibrillation

Ventricular fibrillation is the most common mechanism of sudden unexpected cardiac arrest in adults. **The sooner the patient is defibrillated**, the greater the chance of successful resuscitation. For each minute that defibrillation is delayed, shock success deteriorates by 7-10% per minute. Send someone to fetch the nearest manual or automated external defibrillator (AED) **immediately**. Healthcare professionals are expected to be familiar with the emergency equipment at their disposal, and to formally check their defibrillator **daily** at the start of each shift.

The moment the defibrillator arrives, **lubricate paddles** with defibrillator conduction gel or place specialized defibrillation pads onto the patient's chest while continuing CPR. **Do NOT use sonar gel or KY jelly** - they do not enhance conduction, and may bum the patient.

Place one paddle to the right of the sternum just below the right clavicle, and the other paddle over the left lower ribs in the mid-axillary line. Look at the ECG tracing on the monitor to assess the rhythm. Use **"Quick-look paddle mode"** if ECG electrodes are not yet connected.

If Ventricular Fibrillation (VF) or pulseless Ventricular Tachycardia (VT) is present, administer **one** unsynchronized DC shock (i.e. ensure that "sync" button is switched OFF) and **immediately** resume CPR starting with chest compressions, for 2 minutes. With a monophasic defibrillator, use 360 Joules. With biphasic defibrillators, use 120-360 Joules according to the specific manufacturer's recommendations. If unknown, then use the maximum available energy setting. **NB:** Do **NOT** check the rhythm or pulse directly after a shock - You need to perfuse the heart with 2 minutes of CPR before reassessing the rhythm.

Connect ECG electrodes while continuing high-quality CPR and actively looking for and correcting **contributing causes** (6 H's and 6 T's). Analyze the rhythm after **every 2 minutes** of CPR:

- If **VF/VT persists**, defibrillate followed immediately by another **2 minutes** of good quality uninterrupted CPR.
- If **asystole (a flat-line) is present**, continue high-quality uninterrupted CPR while checking that the **electrode disc** contact is optimal and that all cable **connections** are firmly in place. Increase the **gain** (ECG size) on the monitor screen, and scan through all views (leads **I, II, III**) to ensure that you are not misinterpreting fine VF or artifact (A simple memory aid - a **'Flat-line Protocol'** - can be memorized by remembering the letters "E-C-G" - check "Electrodes - Connections - Gain" on all 3 leads!).
- If an **organized rhythm (Pulseless Electrical Activity) appears** on the monitor, do **NOT** stop chest compressions until the end of the 2 minute cycle (unless the patient has started moving/breathing adequately or the exhaled pCO₂ or arterial pressures indicate a return of pulse), then feel for a central pulse for up to 10 seconds. If a definite pulse is present, continue with post-resuscitation care. If no definite pulse is felt, recommence high quality CPR and reanalyze the rhythm every 2 minutes.

ADVANCED LIFE SUPPORT

While minimizing interruptions to chest compressions:

- Consider advanced airway
- Continuous chest compressions after advanced airway in place
- Consider capnography and arterial monitoring
- Obtain intravenous/intraosseous access
- Consider vasopressors and antiarrhythmics
- Correct contributory causes
- Consider Advanced Airway**

Intubate the trachea only if competent to do so and all the equipment is ready and in good working order. Insertion of a **Laryngeal Mask Airway (LMA) Laryngeal Tube** or **Combitube / Easy tube** is perfectly acceptable as an alternative to tracheal intubation. Continue CPR whilst preparing airway equipment.

When intubation equipment is ready and the laryngoscope and tracheal tube is in the patient's mouth, only then cease chest compressions temporarily to insert the tube through the vocal cords. Once the tube has visibly passed through the cords, **immediately** recommence chest compressions (If more than one attempt is required, perform adequate CPR **between** attempts. Do **not** take more than a few seconds per attempt).

Inflate the tracheal tube cuff with 5ml of air initially, check tube placement with an Oesophageal Detector Device (ODD) if available, then listen for breath sounds bilaterally, listen for absence of epigastric sounds, look for visible chest rise and look for humidification of the tracheal tube. Attach an end-tidal CO₂ device (colorimetric, digital or preferably waveform capnography) if available to **confirm** tracheal placement of the tube. Secure tube in place and **reconfirm** tube placement again. Maintain cuff pressure at <30 cmH₂O. If no cuff pressure monitor is available, listen carefully at the patient's mouth for the presence of an air leak whilst ventilating the patient. If necessary, deflate the cuff until an air leak is just heard, then inflate cuff using 1ml at a time just until the air leak ceases. Minimize interruptions in chest compressions during this confirmation procedure.

- Continuous Chest Compressions after Advanced Airway in Place**

Once an advanced airway device (tracheal tube, LMA or Combitube / Easytube) has been successfully inserted, continuous uninterrupted compressions are performed, with an interposed ventilation given once every **6 seconds**. Do **NOT** hyperventilate!

(iii) **Consider Capnography and Arterial Monitoring**

Quantitative waveform capnography is regarded as the gold standard for confirming initial and continued correct tube placement during resuscitation. Inadvertent and unrecognized displacement of the tracheal tube into the oesophagus is universally fatal.

Waveform capnography is also particularly useful for assessing the quality of CPR performance. A $P_{ET}CO_2$ level of < 10 mmHg suggests **poor quality chest compressions**, and a sudden rise of $P_{ET}CO_2 > 40$ mmHg suggests a possible **return of spontaneous circulation**. Every effort should be made to ensure that waveform capnography is available at all intubation and resuscitation attempts.

(iv) **Obtain Intravenous / Intraosseous Access**

Insertion of an intravenous line, if not already in place, should be attempted **while** chest compressions are continuing. If intravenous access is not readily obtainable, immediate intraosseous access is recommended for both children and adults. Bone marrow aspirate needles and numerous commercial intraosseous devices are readily available, allowing intravascular access to be obtained within seconds. Tracheal administration of drugs is no longer recommended.

(v) **Consider Vasopressors and Antiarrhythmics**

- **Adrenaline** (Epinephrine) - indicated in cardiac arrest not responding to initial resuscitation or defibrillation. Give 1mg (1 ml of 1:1000 solution, diluted with 9 ml sterile water or saline) IV/IO followed by a 20ml sterile water/saline flush after the 2nd shock (if the rhythm is VF/VT), or as soon as intravascular access is obtained (if the rhythm is PEA/asystole). Repeat adrenaline 1mg IV every 3-5 minutes of CPR if cardiac arrest persists. Elevate the limb for at least 10 seconds after each drug administration.
- **Amiodarone** - 300mg IV/IO (diluted to 20ml with 5% D/W) is given after the first dose of Adrenaline (i.e after the 3rd shock) if VF/VT persists. One additional bolus of 150mg may be given after 3-5 minutes if VF/VT still present. After return of spontaneous circulation, a loading infusion of 360mg may be administered over 6 hours at a rate of 1 mg/minute. Thereafter, a maintenance infusion of 540 mg is administered over 18 hours at a rate of 0,5mg/minute. (Maximum total dose - 2,2g/24hrs).
- **Lignocaine** - 1,5mg/kg IV/IO initially, followed by 0,5mg/kg if necessary (Maximum total dose - 3mg/kg). Lignocaine may be given as an alternative to Amiodarone, particularly if Amiodarone is not available.
- **Magnesium** - 1 - 2g IV/IO (diluted to 10ml with 5% D/W or N/S) is given slowly if Torsades de Pointes or hypomagnesaemia is suspected (e.g alcoholic/malnourished), or if the ventricular arrhythmia is due to digitalis toxicity.

(vi) **Correct Contributory Causes**

An aggressive search for reversible or contributory causes of the cardiac arrest must be performed during and after successful resuscitation. These can be conveniently memorized as 6 “H”s and 6 “T”s:

Hypoxia	Tension pneumothorax
Hypovolaemia	Tamponade (Cardiac)
Hypothermia	Toxins and drug overdoses
Hydrogen ion excess (Acidosis)	Trauma
Hypo- / Hyperkalaemia	Thrombosis (Coronary)
Hypo- / Hyperglycaemia	Thrombosis (Pulmonary)

NB:

- **Dilated pupils** may be due to drugs (e.g adrenaline, atropine, antihistamines, tricyclic antidepressants, stimulant abuse, etc), hypothermia, snakebite etc, and therefore does NOT necessarily indicate irreversible brain damage.

Atropine - Administer 0,5 - 1mg IV/IO every 3-5 minutes (up to a total of 3mg) only if a vagal-induced bradycardia is present (Much higher doses may be required for organophosphate poisoning).

Bicarbonate - Administer 1 ml/kg of 8,5% solution slowly IV/IO **only** if severe hyperkalaemia, drug overdosage (tricyclic antidepressants / aspirin / cocaine), or if severe metabolic acidosis is present. Ensure that the patient is intubated prior to sodium bicarbonate administration. Do **NOT** mix with calcium-containing solutions (e.g Ringers Lactate).

Calcium Chloride - Administer 10ml of 10% solution slowly IV/IO **only** with severe hyperkalaemia or beta blocker or calcium channel blocker overdose. Do **NOT** mix with sodium bicarbonate or Balsol.

Dextrose - Administer 50ml of 50% dextrose solution IV/IO slowly **only** if severe hypoglycaemia or hyperkalaemia present.

Insulin - Administer 10 units IV/IO soluble insulin for severe hyperkalaemia (once the 50ml dextrose has been given).

Potassium - If serum potassium levels are low, replace potassium by slowly infusing 20ml (40mEq) KCl over 1 hour. Rapid potassium replacement is lethal!

IMMEDIATE POST-CARDIAC ARREST MONITORING AND SUPPORT

A - Airway Management:

- Maintain and support an open airway.
- Intubate if required.
- Use capnography when available.

B - Breathing Support:

- Optimize **oxygenation and ventilation**. Recheck position of airway adjuncts, take arterial blood gas and ensure correction of hypoxia and hypercarbia.
- If patient does not have adequate spontaneous breathing, intubate and ventilate at 10 breaths/minute initially, titrating PETCO₂ to a target of 35 - 45 mmHg.
- Maintain oxygen saturation at 94 - 98% (but **not** 100%). Consider chest X-ray.
- Apply protective lung ventilation when appropriate (Suggested initial ventilation tidal volume of 6ml/kg and PEEP of > 5 cmH₂O. Target a pH of > 7,2)

C - Circulatory Control:

- Ensure adequate **perfusion**. Correct hypovolaemia and consider inotropic support if hypotensive (e.g. start Adrenaline 0,05 ug/kg/min and titrate to effect). Initially target a systolic BP of > 90 mmHg (MAP of > 65 mmHg).
- Monitor HR, BP, capillary refill, urine output and lactate levels

D - Differential Diagnosis:

- Correct any evident underlying **contributory causes** (H's and T's) and pre-existing pre-arrest medical conditions wherever possible.

E - Evaluation:

- Obtain a **12-lead ECG** tracing (including a right-sided ECG).
- Coronary angiography if the arrest of suspected cardiac origin. Consider **early reperfusion** strategies if evidence of ischaemia (especially STEMI or LBBB).
- Continuous ECG and haemodynamic monitoring (vital signs, SpO₂, end-tidal CO₂, blood glucose, electrolytes, temperature).
- Insert nasogastric and urinary catheters if appropriate.

F - “Freezing” (Targeted Temperature Management) if patient is unresponsive:

- Use established cooling strategies and existing protocols for patients with return of spontaneous circulation who remain comatose (especially if post-VF/VT).
- Maintain a constant targeted temperature between 32 - 36°C for at least 24 hours.
- Monitor glucose, electrolytes (especially K, Ca, Mg, PO₄) and haemodynamic status.
- Rewarm at 0,25 °C per hour. Avoid rebound hyperthermia (Temp > 37,5°C).

G - Glucose Control:

- Maintain blood glucose levels at 8-10 mmol/l
- Avoid hypoglycaemia

H - Head / Neuro Evaluation:

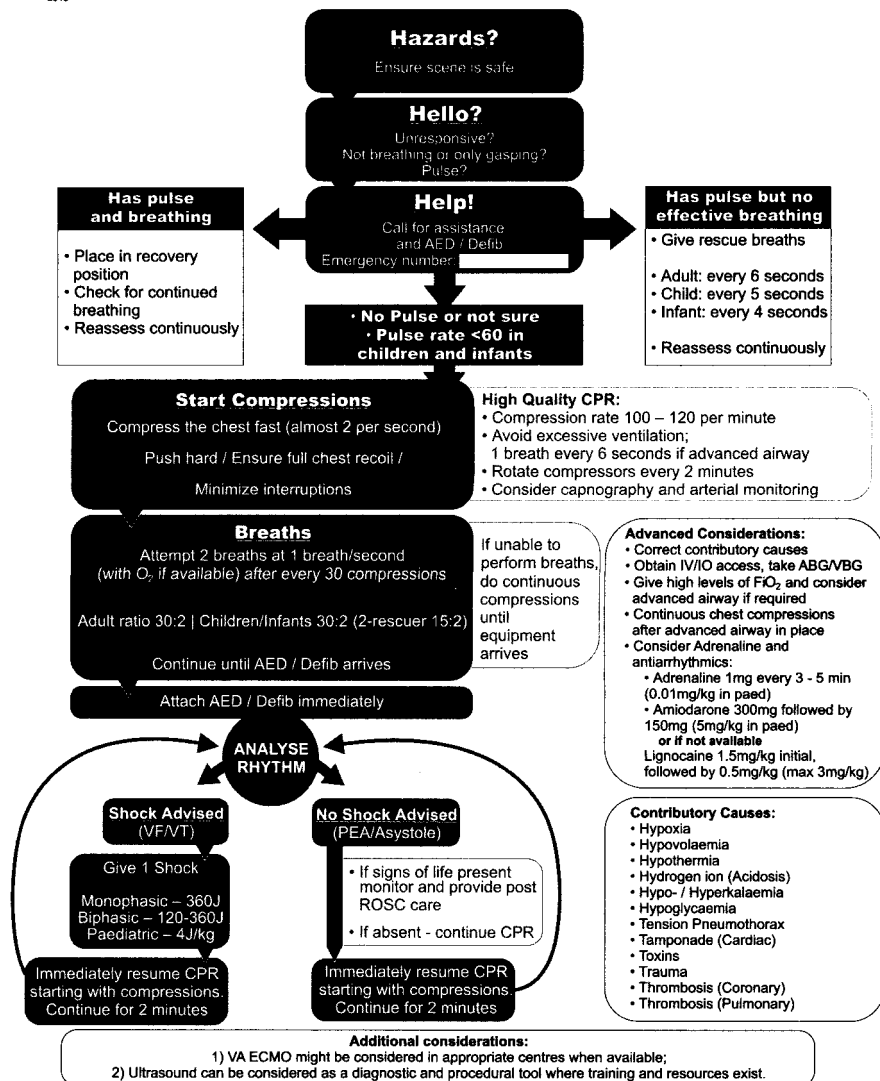
- Treat seizures aggressively. Consider EEG monitoring.
- Consider brain imaging
- Delay prognostication for at least 72 hours post normothermia

Consult relevant specialist colleagues for further post-resuscitation critical care management. Whenever appropriate, arrange transfer of the patient to a specialist unit by appropriate mode of transfer.

Contact, communicate with and update regularly patient’s family or guardians throughout the resuscitation process wherever possible. Consider post-resuscitation debriefing for family and resuscitating personnel as soon as possible.

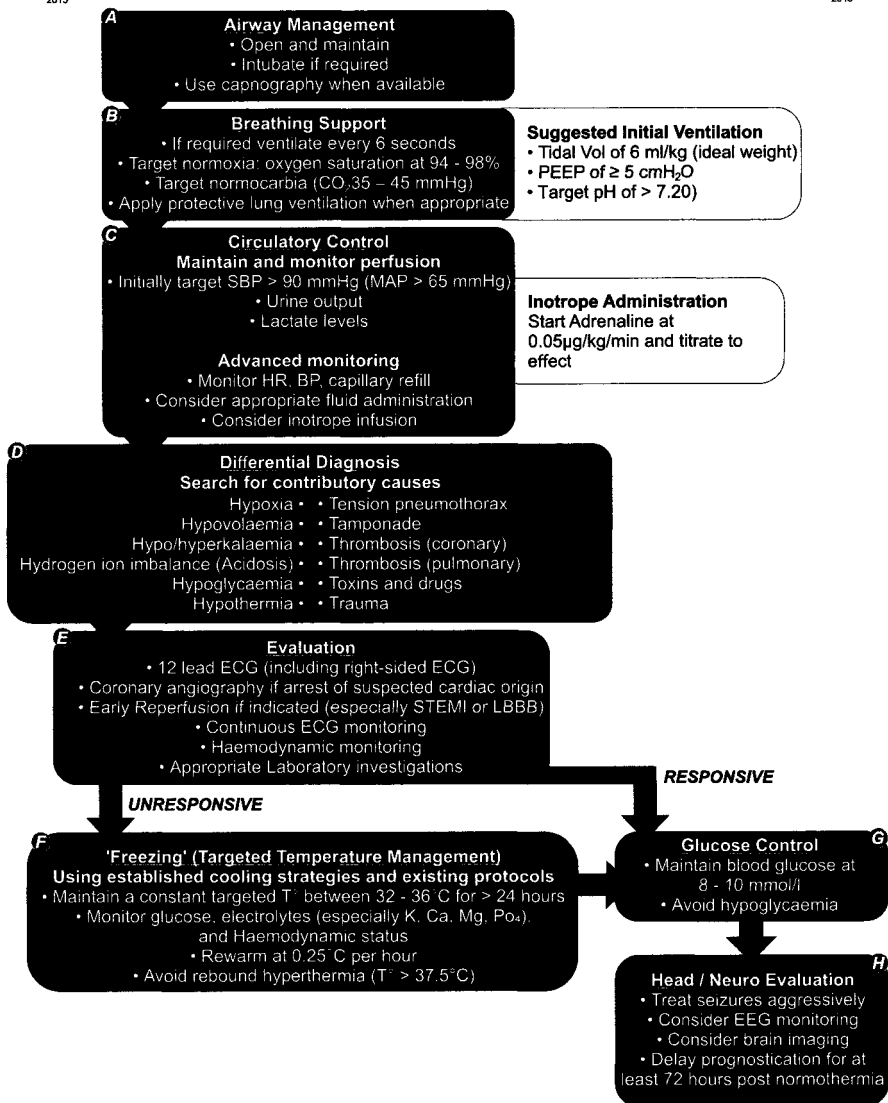
The accompanying **Advanced Cardiac Arrest Algorithm and Post Cardiac Arrest Care Algorithm**, based on the recommendations of the **International Liaison Committee on Resuscitation (ILCOR)**, provides a complementary approach.

ADVANCED CARDIAC ARREST ALGORITHM Adult and Paediatric



POST CARDIAC ARREST CARE

(Return of Spontaneous Circulation)



SECTION 12 ACUTE LARYNGEAL OR TRACHEAL OBSTRUCTION

SIGNS AND SYMPTOMS

- (1) Stridor
- (2) Dyspnoea
- (3) Restlessness
- (4) Vigorous use of accessory muscles of respiration
- (5) Marked distension and emptying of jugular veins with expiration and inspiration respectively.
- (6) Intercostal recession
- (7) Cyanosis
- (8) Collapse

CAUSES

- (1) Epiglottitis, laryngo-tracheo-bronchitis (more common in infants and children).
- (2) Foreign body (e.g. food, sweets, beads etc).
- (3) Laryngeal oedema, anaphylaxis.
- (4) Unconscious / weak patients - tongue or secretions or aspiration.
- (5) Blocked tracheal tube - due to kinking or blood clot or encrusted sputum.
- (6) Trauma.

TREATMENT OF FOREIGN BODY AIRWAY OBSTRUCTION

(i) Patient Conscious

- Encourage patient to **cough** if possible, and watch for improvement or deterioration.
- If patient unable to cough, talk or breathe, administer up to 5 **abdominal thrusts** (Heimlich Manoeuvre) - stand behind the patient and wrap your arms around patient's waist, placing your fist (one hand on top of the other) in the midline just above the patient's navel, and administer a series of up to 5 quick inward and upward thrusts (Can be self-administered if you happen to be the victim!)
- If the 5 abdominal thrusts are unsuccessful, lean patient forward and administer up to 5 **back slaps**.
- Repeat abdominal thrusts and back slaps until object dislodged or patient becomes unconscious.

(ii) Patient Grossly Obese or Advanced Pregnancy

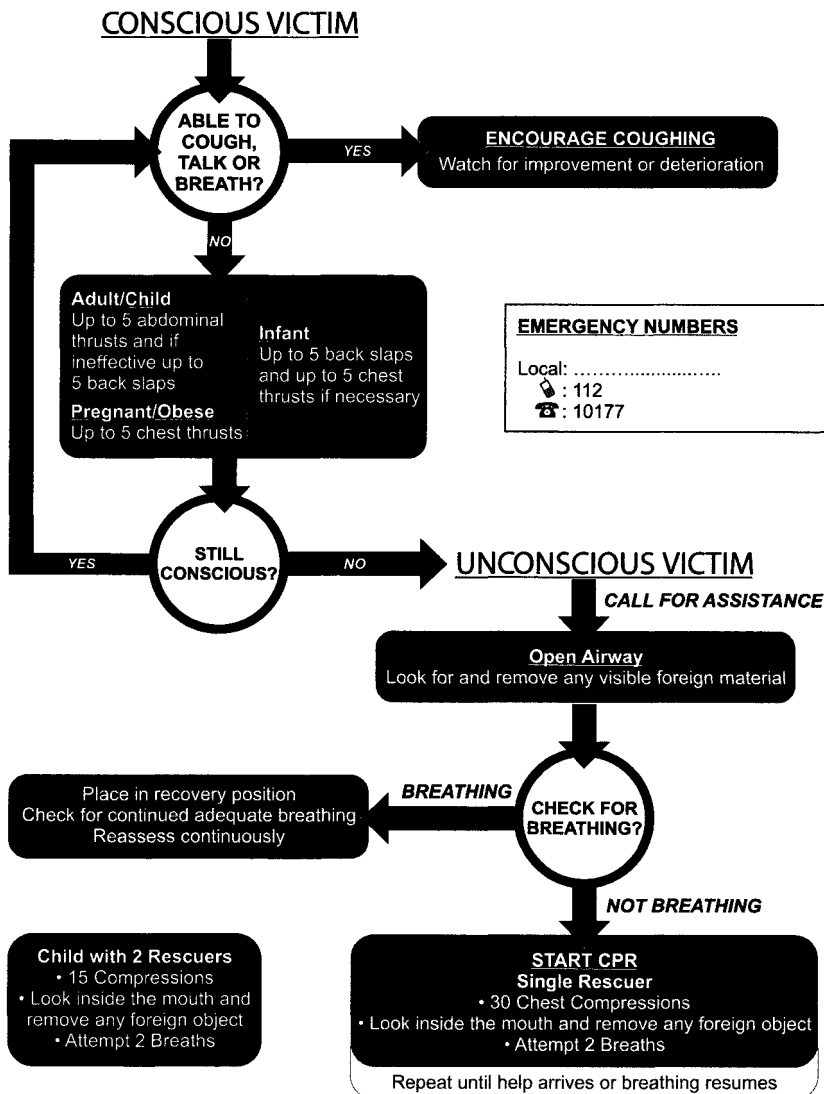
- Place your arms around patient's chest under the armpits, forming a fist over the mid- sternum, and apply up to 5 quick backward-directed chest thrusts.
- If unsuccessful, attempt up to 5 back slaps. Repeat chest thrusts and back slaps until object dislodged or patient becomes unconscious.

(iii) Patient Unconscious

- Immediately call for assistance and place patient supine. Open the airway (**head tilt- chin lift**) and look for any visible object in the mouth.
- If visible, carefully remove object under **direct visualization** using a Magills forceps (if available) or a **Finger Sweep**; grab patient's tongue and jaw upwards with the thumb and fingers of one hand while inserting the index finger of your other hand inside along the patient's cheek down to the base of the tongue and remove the foreign body with a hooking action (**Be very careful** not to push it further down!).
- Assess if the patient is breathing. If not breathing, start CPR by giving 2 breaths after every 30 chest compressions.
- Open the airway and look for any visible foreign material after every 30 compressions. Repeat the sequence of 2 breaths and 30 compressions until patient responds or assistance arrives.
- If equipment available, perform direct laryngoscopy and tracheal intubation or emergency needle or surgical cricothyroidotomy if necessary.

The accompanying **Choking Algorithm** and **Advanced Airway Management Algorithm** provides a complementary approach.

CHOKING ALGORITHM



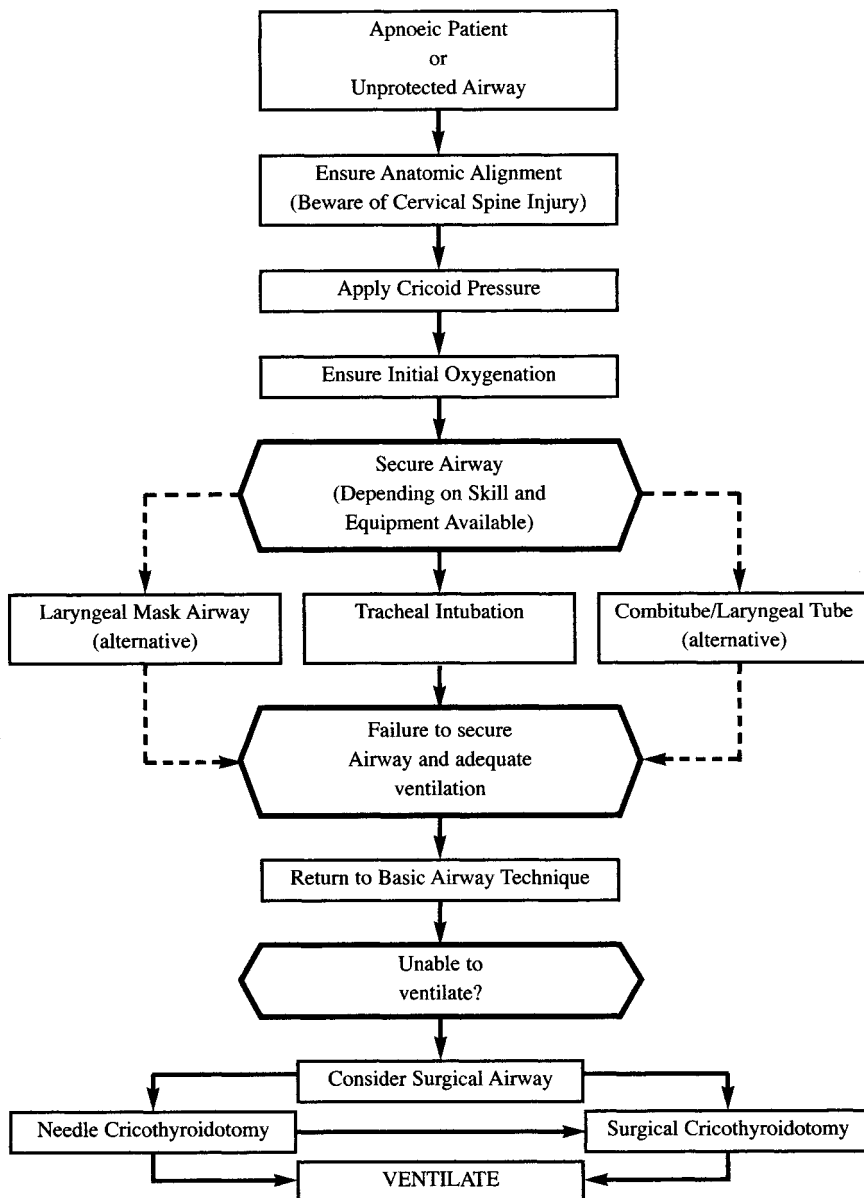
EMERGENCY NUMBERS

Local:

☎ : 112

☎ : 10177

Advanced Airway Management Algorithm (Adult and Child)



SECTION 13 NEAR DROWNING

PRE-HOSPITAL MANAGEMENT

1. Get to victim **as quickly and safely as possible** (preferably with some flotation device).
2. Start rescue breathing (mouth-to-mouth / mouth-to-nose / mouth-to-snorkel) **as you get to the victim in the water** or as soon as this can safely be done.
3. If neck injury suspected, keep head neutral (neither flexed nor extended), float victim supine onto a horizontal back support before removing from the water if possible, and open the airway using the jaw-thrust or chin lift manoeuvre **without** head tilt.
4. **DO NOT** try to remove water from the lungs. Just suction or wipe mouth clear as necessary by wrapping a handkerchief or cloth around your fingers, or rolling the victim onto his/her side, and continue rescue breathing. (Often very little water actually gets into the lungs).
5. Start chest compressions once the victim is horizontal on a firm, flat surface and pulse is found to be absent.
6. High risk of aspiration pneumonitis during CPR because of the large volumes of water **swallowed** during near drowning. Therefore, **all** victims require immediate transfer and admission to hospital, no matter how quickly the victim responded to CPR.
7. Give oxygen and monitor continuously en route to hospital. **Pat victim dry** (do not **rub**).

NB: Successful resuscitations with complete recovery can occur even after prolonged cold water submersion and fixed dilated pupils. Therefore start CPR **regardless** of how long victim has been submerged, and continue **uninterruptedly** while rewarming the victim. Do not stop CPR until patient is **normothermic** (> 35°C) with no evidence of brain function (Brain function tests are **unreliable** in hypothermic patients).

HOSPITAL MANAGEMENT

1. Continue CPR if necessary.
2. Intubate if indicated.
3. Check blood gases and chest X-ray and watch out for the development of pulmonary oedema post-resuscitation. Ventilate with CPAP or PEEP post-resuscitation if necessary.
4. Pass a nasogastric tube and aspirate gastric contents.
5. Monitor BP, ECG, temperature, urine output and serum electrolytes (insert CVP line and give fluids / inotropes if necessary).
6. Rewarm victim.
7. Give antibiotics if indicated.
8. Steroids are of no proven value (and may be detrimental).
9. Sodium bicarbonate is not usually indicated, as metabolic acidosis will improve with re warming and improved circulation.

COMPLICATIONS OF NEAR DROWNING

1. Pulmonary - aspiration / pulmonary oedema (early or delayed onset).
2. Cardiac - arrhythmias / hypotension / myocardial depression.
3. CNS - hypoxic brain damage / convulsions
4. Renal - renal failure.
5. Metabolic - acidosis / electrolyte abnormalities.
6. GIT - gastric distension / foul diarrhoea.
7. Haematological - haemolysis / DIC.

NB: Always ask "Why did the drowning occur?" The victim may have lost consciousness for some or other reason (e.g. epilepsy, alcohol, drug overdose, spinal injury, acute myocardial infarction etc) - manage appropriately.

SECTION 1.4 ACUTE CORONARY SYNDROMES

Acute coronary syndromes are divided into:

- (A) **UNSTABLE ANGINA**
- (B) **NSTEMI** - Non-ST-Elevation Myocardial Infarction.
- (C) **STEMI** - ST-Elevation Myocardial Infarction (classical Myocardial Infarction).

(A) **UNSTABLE ANGINA**

- Angina which occurs at rest or that is increasing in frequency and/or severity.
- Troponin testing is negative.

(B) **NON ST-ELEVATION MYOCARDIAL INFARCTION**

1. **DEFINITION**

- Angina that is associated with typical rise/fall of cardiac biomarkers and ST- segment depression or T-wave inversion.

2. **MANAGEMENT PLAN**

- Admit patient, preferably to a high care unit or Coronary Care Unit.
- Take bloods for cardiac markers and Troponin T or I (has prognostic value).
- Treat aggravating factors (eg. uncontrolled hypertension, cardiac failure, arrhythmias, infection, anaemia).

Risk Stratify the Patient

TIMI Risk Score (each risk factor scores 1 point)

1. Age > 65 years
 2. >_3 coronary artery risk factors
 3. Known coronary artery stenosis > 50%
 4. Aspirin use in the last 7 days
 5. Elevated cardiac biomarkers
 6. Severe angina (> 2 episodes in < 24 hours)
 7. ST-depression or elevation > 0,5mm
- (0 - 2 = Low Risk 3 - 4 = Medium Risk 5 - 7 = High Risk)

Drug Management

1. **Aspirin:** 300mg stat (chewed) and then 150mg daily (unless allergic to aspirin). Do not use enteric-coated aspirins. Use with caution in patients with asthma or bleeding disorders
2. **Clopidogrel:** 300 - 600mg stat orally and then 75mg daily.
Alternatives are Prasugrel or Ticagrelor
3. **Anticoagulation:** Either unfractionated heparin IV (keep PTT 2x normal) or low molecular weight heparin subcutaneously (Enoxaparin 1 mg/kg b.d.). Anticoagulate for at least 48 hours.
4. **Beta blockers:** If no contraindication is present.
5. **Statins:** High dose statin therapy should be started as soon as possible (e.g. Atorvastatin 80 mg daily).
6. **Nitrates:** For pain control. Use with extreme caution if there is hypotension or hypo- volaemia (or sexual enhancement drug usage such as sildenafil (Viagra) etc), or if the patient has bradycardia (<50/minute) or tachycardia (> 110/minute).
7. **Morphine:** Must be used with caution in patients with NSTEMI (as it has been associated with increased mortality in some cases)
8. Aim to stabilize patient. If chest pain settles and patient low risk, plan for an exercise ECG in 2-4 weeks.
9. If patient unstable (ongoing chest pain, cardiac failure) refer urgently (within 24 hours) for coronary angiography and subsequent revascularization.

3. NOTE:

- Early aspirin reduces mortality significantly.
- Beta blockers in unstable angina reduces acute myocardial infarction by 13%.
- Heparin decreases acute myocardial infarction by 8-15%.
- Calcium antagonists in unstable angina are controversial and are used mainly for coronary spasm.

(C) ST-ELEVATION MYOCARDIAL INFARCTION (CLASSICAL MI)

1. CONFIRM DIAGNOSIS

- Typical history
- Clinically: Distant heart sounds, S3 gallop, atypical precordial impulse, appearance or accentuation of murmurs with chest pain.
- ECG changes (12 - lead ECG, or preferably 16-lead ECG including V4R)
- Elevated cardiac bio markers

2. EXCLUDE OTHER LIFE-THREATENING CAUSES OF CHEST PAIN

- Pericarditis / Cardiac tamponade
- Dissecting aortic aneurysm
- Pulmonary embolus
- Tension pneumothorax

3. RISK STRATIFY THE PATIENT

TIMI Risk Score

Age >75 years

Age 65 -74

Diabetes Mellitus or Hypertension or Angina

SBP <100 mmHg

HR > 100/min

Killip Class II to IV

Weight > 67 kg

Anterior STEMI or LBBB

Time to reperfusion > 4 hours

(0 - 2 = Low Risk 3 - 4 = Medium Risk 5 - 7 = High risk)

4. MANAGEMENT Admit to

ICU for monitoring

a. General measures

- (i) Bed rest in semi-Fowler's position (unless hypotensive or comatose)
- (ii) Oxygen via nasal cannula or mask to keep sats at 94-99% (but not 100%).
- (iii) Establish IV line (200ml N/S).
- (iv) Continuous ECG monitoring and monitoring of vital signs.

b. Measures to limit myocardial infarct size and decrease mortality

- (i) **Aspirin:** Single most cost-effective therapy for MI. It decreases mortality significantly. Give 300mg to chew immediately to facilitate rapid absorption, followed by 150mg (unless allergic to aspirin). Do **not** use enteric-coated aspirins. Use with caution in patients with asthma or bleeding disorders
- (ii) **Clopidogrel:** 300 - 600mg stat orally and then 75mg daily.
Alternatives are Prasugrel or Ticagrelor

- (iii) **Statins:** High dose statin therapy, started as soon as possible, has been shown to be beneficial (e.g. Atorvastatin 80 mg).
- (iv) **R Blockers:** Should be initiated within the first 24 hours after the onset of symptoms for all patients without contraindications such as heart failure, low output state, increased risk of cardiogenic shock, heart block, asthma or significant COPD. Avoid IV β blockers early after infarction.
- (v) **ACE Inhibitors:** Early use of ACE Inhibitors, within 24 hours of infarction, is beneficial in decreasing mortality especially in large infarcts. Administer only once the patient is haemodynamically stable.
- (vi) **Calcium Antagonists:** To date there is no evidence to support the routine use of calcium antagonists in the acute phase of myocardial infarction.

c. Measures to control cardiac pain

- (i) **Nitrates:** Contraindicated if hypotension (SBP < 90mmHg or > 30mmHg below baseline) is present. Ensure patient has not had phosphodiesterase inhibitors such as sildenafil (Viagra) or vardenafil (Levitra) within 24 hours, or tadalafil (Cialis) within 48 hours. Contra-indicated in RV infarction or if the patient has bradycardia (<50/minute) or tachycardia (>110/minute).
Sublingual - 0,5mg tablet or 0,4mg spray at 5 minute intervals if necessary, up to a maximum of 3 doses. Check blood pressure and heart rate before each dose. Do **not** shake the spray, and ensure that the patient holds his/her breath briefly while administering the spray.
IV infusion: Start at 10ug/min and titrate to effect, increasing by 10ug/min every 3-5 minutes if necessary. Aim to decrease systolic BP by no more than 10% (if normotensive) or 30% (if hypertensive). SBP must not go below 90 mmHg. (Maximum ceiling dose - 200ug/min).
- (ii) **Morphine:** If pain persists after nitrate administration and patient not hypotensive or hypovolaemic. Prepare by diluting 15mg with 14ml of water. Administer 2 - 4mg slowly IV at 5 - 15 minute intervals, just until pain relief is obtained or side effects (hypotension, nausea, vomiting, respiratory depression) appear. Use morphine with extreme caution in RV infarction.

d. Reperfusion Measures

- (i) **Primary Percutaneous Coronary Intervention (PCI):** First choice. Has a small but significant mortality benefit as compared with fibrinolytic therapy. Benefit is seen particularly in patients undergoing **primary PCI within 120 minutes of first medical contact** in centres which are highly experienced with these procedures.
- (ii) **Fibrinolytic Therapy:** Fibrinolytic agents have shown a significant reduction in mortality and should be used in **all eligible patients who are not able to undergo timely PCI**. Fibrinolysis is of most benefit if given within 6 hours, but may still be beneficial up to 12 hours after onset of chest pain. *. In many developing countries, including South Africa, timely PCI is not possible in majority of cases and thus early fibrinolysis should be the preferred first choice of therapy for these patients*
- (iii) **Patients who are shocked:** Start inotropic support with dobutamine and immediate transfer to cardiac cath lab for coronary angiography / revascularization and haemodynamic support with intra-aortic balloon counterpulsation.

4. PROPHYLACTIC ANTI-ARRHYTHMICS

Lignocaine has been shown to decrease the incidence of ventricular fibrillation in some studies, however, none of the prophylactic anti-arrhythmic drugs have reduced mortality. Thus anti-arrhythmic drugs should ideally only be given for haemodynamic compromising arrhythmias.

5. PROTOCOL FOR FIBRINOLYTIC ADMINISTRATION

(i) Absolute Contraindications

- Previous intracranial haemorrhage or stroke of unknown origin at any time
- Ischaemic stroke within last 6 months
- Cerebral vascular lesion (e.g. AV malformation)
- Intracranial malignancy
- Recent major trauma/surgery/head injury < 3 weeks
- Gastrointestinal bleeding within past month
- Dissecting aortic aneurysm
- Active bleeding or bleeding diathesis (excluding menstruation)
- Non-compressible punctures in past 24 hours (e.g. liver biopsy, lumbar punctures)

(ii) Relative Contraindications

- Chronic severe poorly controlled hypertension
- Refractory hypertension (SBP >180 mmHg or DBP >110 mmHg)
- Transient ischaemic stroke < 6 months ago
- Traumatic or prolonged CPR
- Major surgery within the previous 3 weeks
- Advanced liver disease
- Active peptic ulcer
- Current use of anticoagulants
- Pregnancy or within 1 week post-partum

(iii) Fibrinolytic Administration

Administer fibrinolytic within 30 minutes of presentation.

Metalyse preferred as it can be given as a bolus dose.

Streptokinase should be used only if metalyse or actilyse not available

	Dose	Contraindications
Tenectoplas (Metalyse)	< 60 Kg : 30 mg IV bolus over 5 secs 60-69 kg : 35 mg IV bolus over 5 secs 70-79 kg : 40 mg IV bolus 80-89 kg : 45 mg IV bolus >90 kg : 50 mg IV bolus	Do not mix with dextrose solutions
Actilyse (rTPA)	15 mg IV bolus, followed by 0,75 mg/kg over 30 minutes (Maximum 50 mg), then 0,5 mg/kg over 60 minutes (Maximum 35 mg) Maximum total dose not to exceed 100 mg.	
Streptokinase	1,5 million Units IV over 60 minutes	Prior streptokinase

Adjunctive anticoagulants with fibrinolytics:

Unfractionated Heparin:

- 60 U/kg IV bolus (Maximum 5000 U), followed by 12 U/kg/hr IV infusion (Maximum 1000 U/hr) for 24 - 48 hrs.
- Check initial aPTT at 3 hrs, then every 6 hrs until stable.
- Target aPTT at 50 - 70 seconds.
- Do not use if platelet count < 100,000.

Low Molecular Weight Heparin (Enoxaparin):

- If < 75 years old - Give 30 mg IV bolus, followed 15 minutes later with 1 mg/kg subcutaneously 12 hourly (Maximum of 100 mg/dose).
- If > 75 years old - No initial IV bolus. Just give 0,75 mg/kg subcutaneously 12 hourly (Maximum of 75 mg/dose).
- If renal insufficiency present (creatinine clearance < 30 ml/minute) - No initial IV bolus. Just give 1 mg/kg subcutaneously every 24 hours.
- Do not use if platelet count < 100,000.

SECTION 1.5 CARDIAC DYSRHYTHMIAS

1. SINUS TACHYCARDIA



Recognition:

Rate > 100/minute. P waves identifiable before each QRS complex.

Management:

Treat the underlying cause, e.g. pain, fever, fear, fright, anxiety, trauma, hypovolaemia, anaemia, cardiac failure etc (The upper limit of physiological tachycardia in adults is usually regarded as 220 minus the patient's age).

2. ATRIAL FIBRILLATION WITH RAPID VENTRICULAR RESPONSE



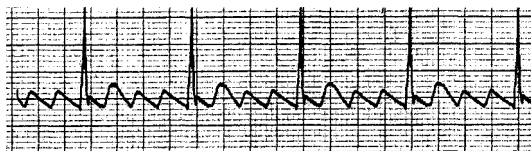
Recognition:

Rate is variable (irregularly irregular QRS intervals). No P waves are identifiable. Wavy baseline.

Management:

(See below under "Atrial Rutter")

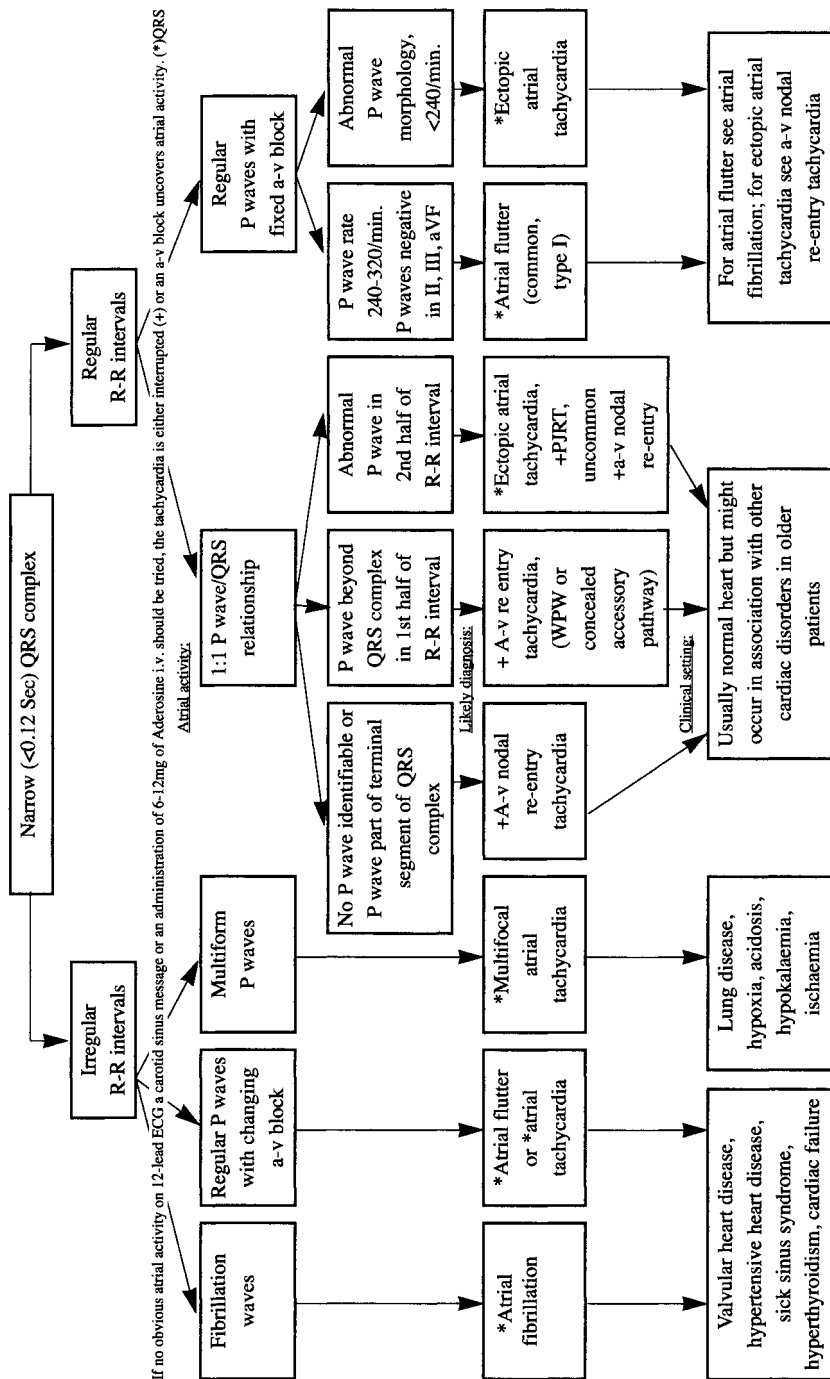
3. ATRIAL FLUTTER



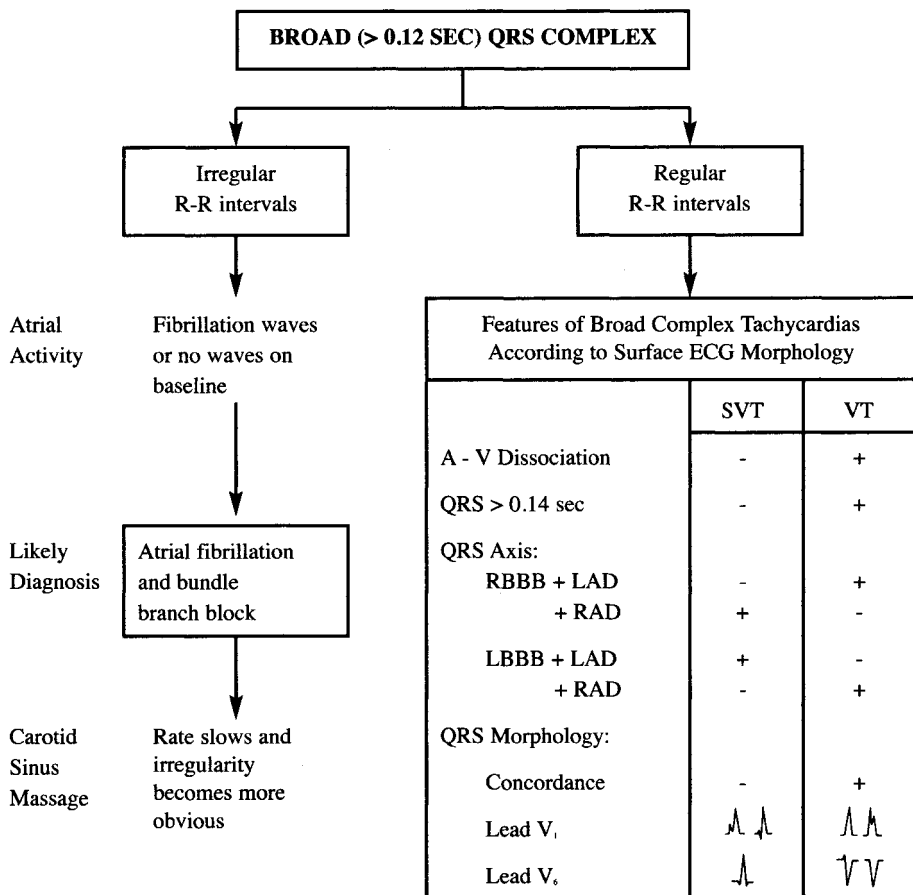
Recognition:

Note presence of P waves with an atrial rate greater than 250/min (often known as the "sawtooth pattern" of Atrial Rutter). The rhythm may be regular or irregular with varying degrees of AV block.

ELECTROCARDIOGRAPHIC DIAGNOSIS OF PAROXYSMAL TACHYCARDIAS



ELECTROCARDIOGRAPHIC DIAGNOSIS OF PAROXYSMAL TACHYCARDIAS (Continued)



Management of Atrial fibrillation and Atrial Flutter:

1. General Principles:

- Search for and correct precipitating factors such as coronary artery disease, hypoxia, anaemia, hypertension, congestive heart failure, pulmonary embolism, thyrotoxicosis, electrolyte abnormalities (low potassium or low magnesium), drug-induced (especially digoxin, quinidine, alcohol or stimulants), valvular disease.
- Expert consultation should always be obtained, particularly with regard to the choice and use of the various pharmacological agents
- If arrhythmia causes significant hypotension or aggravates cardiac failure or angina, synchronized cardioversion should be considered. Start with 100J in Atrial Flutter and with 200J in Atrial fibrillation (monophasic or biphasic defibrillation). Give an IV bolus of heparin (60U/kg) beforehand and follow with anticoagulation for at least 4 weeks if the arrhythmia was chronic (> 48 hours)
- If 3 attempts at cardioversion are unsuccessful, consider amiodarone (300mg over 10 - 20 minutes, and repeat the cardioversion. Follow with an amiodarone infusion (900mg over 24 hours).
- If the patient is stable and the arrhythmia is acute (< 48 hours) - aim for restoration of sinus rhythm.
- If the patient is stable and the arrhythmia is chronic (> 48 hours) - aim to control ventricular rate.

2. Sinus Rhythm Restoration (under Expert Supervision)

- Pharmacologic cardioversion:
 - Amiodarone or sotalol may be considered in the presence of cardiac dysfunction.
 - Flecainide or propafenone may be considered in the **absence of organic heart disease**.
- Elective cardioversion (preferable):
 - In stable chronic Atrial Flutter / fibrillation (> 48 hours) an intracardiac thrombus must be excluded and patients should ideally be anticoagulated for 3 weeks before and continued for 4 weeks after cardioversion).

3. Ventricular Rate Control (under Expert Supervision)

- Beta blockers or calcium channel blockers can be used for rate control if there is no organic heart disease.
- Digoxin can be considered in the presence of impaired cardiac function.
- **NB:** Concomitant use of beta blockers and calcium channel blockers should be **avoided**. Digoxin levels are increased by verapamil and amiodarone.

4. Atrial fibrillation associated with Wide QRS Complexes

(Pre-excited Atrial fibrillation)

- **Avoid** all AV nodal blocking agents such as adenosine, beta blockers, calcium channel blockers and digoxin.
- Synchronized cardioversion is the treatment of choice if the patient is unstable.
- Amiodarone can be considered if cardioversion unsuccessful.

4. NARROW COMPLEX TACHYCARDIA (SUPRAVENTRICULAR TACHYCARDIA)



Recognition:

Rate usually > 150/minute with narrow QRS complexes (< 0,12sec).

Sudden onset with no obvious physiological cause. P-waves seldom seen.

Management:

A - Airway: Open, maintain and protect as necessary.

B - Breathing: Administer oxygen. Ventilate if necessary. Target sats 94-98%.

C - Circulation: Assess pulse, blood pressure and perfusion. Attach ECG monitor, pulse oximeter and vital signs monitor if available.

D - Drip: Establish an IV line with Normal Saline.

E - ECG Rhythm: Run rhythm strip to confirm dysrhythmia. Do a 12-lead ECG if possible.

Identify and treat any underlying causes. Specialist medical advice should be sought whenever possible.

Management of the Stable Patient

1. Vagal Stimulation:

May be attempted if the patient is stable and the heart rate is **regular** (no varying R-R intervals suggesting atrial fibrillation). Watch BP and heart rate and record rhythm strip while performing these procedures. Effectiveness may be enhanced by combining the manoeuvres

(a) Valsalva Manoeuvres (forced expiration against a closed glottis):

- Deep repetitive forceful coughing.
- Place your hand on the patient's abdomen and ask patient to forcefully push your hand away for at least 15 seconds.
- Ask patient to hold their breath and strain forcefully for at least 15 seconds.
- Modified Valsalva Manoeuvre: Ask patient in a sitting position to blow into the tubing of a BP manometer, keeping the pressure at 30mmHg for at least 15 seconds. Then lay patient flat on the bed and raise both legs to 30° for 15 seconds

(b) Facial Application of Ice Water:

- Place ice (wrapped in a **wet** cloth) onto the patient's forehead (but not over the patient's eyes) for at least 15 seconds.

(a) Carotid Sinus Massage:

- May be attempted if no underlying vascular disease is present:
 - Age - Avoid in elderly patients
 - Bruits - Ensure absence of carotid bruits bilaterally
 - Cholesterol - Look out for evidence of high cholesterol levels
 - Disease of the CVS (Cardiac, cerebral and / or peripheral)
 - ECG - Ensure no previous or silent infarct (e.g. Q waves)
 - Feel carotid pulsations bilaterally - Ensure equal strength.
- Turn patient's head to the left and massage the right carotid sinus (located just above the thyroid cartilage) longitudinally with firm posteromedial pressure for 5 seconds. Pause for 10 seconds and repeat twice more if necessary.
- If 3 attempts on the right carotid sinus have been unsuccessful, one can attempt left carotid sinus massage up to 3 times, pausing for 10 seconds between each attempt.
- **Never** perform simultaneous bilateral carotid sinus massage!

1. Adenosine:

- May be administered if the patient is stable and the heart rate is regular (no varying R-R intervals suggesting atrial fibrillation) and vagal stimulation techniques have been unsuccessful.
 - Watch BP and heart rate and **record rhythm strip** while administering adenosine. Transient bradycardia, ectopy or asystole may occur.
 - 6mg adenosine is given as an **extremely rapid** IV bolus through a large bore IV cannula, followed **immediately** by a 20ml N/S flush and lifting up the arm. Ideally use a 3-way stopcock or multiple injection port close to the patient, so that there is no delay between administering the adenosine and the flush.
 - A second dose, using 12mg adenosine may be given 1-2 minutes later if the 6mg dose was ineffective.
 - Transient side-effects include flushing, chest tightness or pain, nausea and / or shortness of breath. Position patient semi-Fowlers and ask patient to cough deeply when symptoms or brief asystole occurs.
 - A lower initial dose (3mg) may be required if the patient is on carbamazepine, dipyridimole, beta blockers or calcium channel blockers, heart transplant patients, or if given through a central line.
 - A higher initial dose (12mg) may be required if the patient is taking theophylline, caffeine or theobromine.
 - Adenosine can be used safely in pregnancy, but is contraindicated in asthmatic patients.
- 2.** If tachycardia repeatedly recurs after termination with adenosine, or if adenosine is unsuccessful or contraindicated, obtain expert opinion with regard to the use of the following:
- Beta blockers
 - Calcium channel blockers
 - Amiodarone (150mg in 5% D/W over 10 min, then 1mg/min x 6hrs, then 0.5mg/min x 18 hrs)
 - Elective synchronized cardioversion.

4. Supraventricular tachycardias are increasingly being cured by radiofrequency ablation of the accessory pathway.

Management of the Unstable Patient

If the patient has a pulse but shows signs of instability such as hypotension, acutely altered mental state, signs of shock, ischaemic chest discomfort and / or acute heart failure, and the signs are caused by the dysrhythmia, the patient will require immediate synchronized cardioversion.

1. Synchronized Cardioversion:

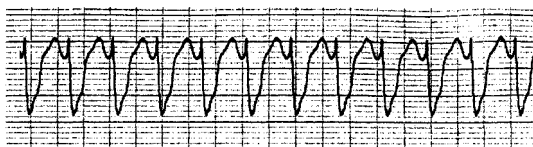
(Consider procedural sedation if patient pain-responsive)

- Ensure that the “sync” button is switched on.
- Locate “sync” marker on monitor screen. If necessary, increase the gain (ECG size).
- Select appropriate energy level - 100J initially (monophasic or biphasic defibrillators).
- Ensuring that all personnel are clear of the patient, and that any oxygen delivery device is at least 1 metre away from the patient, charge the defibrillator and aim to shock the patient on expiration.
- Press the discharge buttons and defibrillator paddles on the patient for at least 2 seconds, until the shock has been delivered (noticeable muscle spasm).
- If the tachycardia persists, repeat above 5 steps (a) to (e), but using the next higher energy level setting (Joules). **Make sure that the “sync” button is ON prior to each cardioversion attempt** (otherwise VF could be induced)!

2. Adenosine:

If the rhythm is **regular** (NOT atrial fibrillation), adenosine (as described above) can be attempted if large bore IV access is readily available, particularly if a defibrillator is not immediately available or if synchronized cardioversion is unsuccessful.

5. MONOMORPHIC WIDE COMPLEX TACHYCARDIA (VENTRICULAR TACHYCARDIA)



Recognition:

Rate usually > 150/minute with wide similar QRS complexes (> 0,12sec). P waves seldom discernable.

Management:

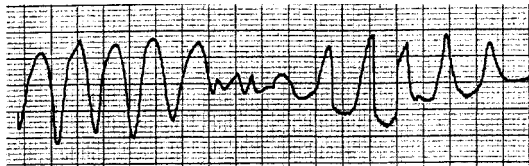
“A - B - C - D - E” as described for Narrow Complex Tachycardia (SVT) above.

Management of the Unstable Patient

If the patient has a pulse but shows signs of instability such as hypotension, acutely altered mental state, signs of shock, ischaemic chest discomfort and / or acute heart failure, and the signs are caused by the dysrhythmia, the patient will require **immediate synchronized cardioversion**, as described for Narrow Complex Tachycardia (SVT) above.

Management of the Stable Patient:

1. **Adenosine:** Can be considered if the rhythm is **regular** and **monomorphic**. Use as described for SVT above. It must **not** be given if the rhythm is irregular or polymorphic (as VF may result!)
 2. **Amiodarone:** Useful for the control of resistant ventricular tachycardia. Administer 150mg over 10 minutes (15mg/min), then 1mg/min for 6 hours, then 0,5mg/min for 18 hours (Maximum dose - 2,2g over 24 hours).
 3. **Lignocaine:** Only to be used if amiodarone is not available, as it is not as effective as amiodarone. Administer 1- 1V₂ mg/kg lignocaine IV, followed by a maintenance infusion of 1 - 4mg/min (30 - 50ug/kg/min).
 4. **Sotalol** (1,5mg/kg IV over 5 minutes) or **Procainamide** (20 - 50mg/min IV, then 1 - 4mg/min infusion) are alternative options in the absence of a prolonged QT interval. These drugs should only be used under expert supervision.
 5. **Implantable defibrillators** are ideal for patients with recurrent VT with hypotension and / or syncope.
6. **POLYMORPHIC (IRREGULAR) WIDE COMPLEX TACHYCARDIA**
(Including Torsades de Pointes)



Recognition:

Rate usually > 250/minute with wide QRS complexes (> 0,12sec) which are bizarre and variable in appearance.

Management:

“A - B - C - D - E” as described for Narrow Complex Tachycardia (SVT) above.

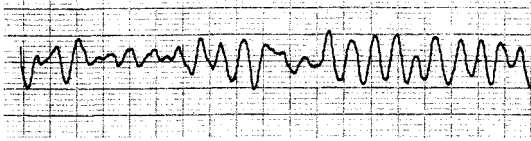
Management of the Unstable Patient (Seek expert consultation):

1. Unstable polymorphic (irregular) VT requires immediate **defibrillation** (not synchronized cardioversion).
2. **Polymorphic VT with a long QT interval (Torsades de Pointes):**
 - **Magnesium sulphate** - 1 - 2g IV (diluted in 50ml of 5%DW) over 10 minutes, followed by an infusion of 0,5 - 1g/hour.
 - **Correct electrolyte imbalance** (low magnesium, low potassium, low calcium).
 - **Discontinue any drugs** that may prolong the QT interval.
 - **Overdrive pacing** (at rates of 100 - 120/min) may be effective for torsades associated with bradycardia and drug-induced QT prolongation.
 - **Lignocaine** - May be considered, but its effectiveness in Torsades has not been established.

3. **Polymorphic VT with a normal QT interval:**

- **Amiodarone** may be effective (Magnesium is of no benefit if the QT interval is normal).
- **Beta blockers** (if associated with myocardial infarction).
- **Isoproterenol** (if associated with Brugada syndrome).

7. **VENTRICULAR FIBRILLATION / PULSELESS VENTRICULAR TACHYCARDIA**



Recognition:

VF is characterized by the complete absence of an organized rhythm.

No pulse detectable.

Management:

1. **IMMEDIATE** defibrillation is the only treatment for VF and pulseless VT. Administer an unsynchronized shock, followed immediately by 2 minutes of CPR. Use 360J (monophasic defibrillators) or according to the manufacturer's recommendations (biphasic defibrillators). Repeat the shock every 2 minutes if VF/VT persists.
2. Administer adrenaline 1mg IV after the 2nd shock, and every 3-5 minutes thereafter if VF/VT persists.
3. Consider amiodarone 300mg IV bolus after the 3rd shock, and a further 150mg after 3- 5 minutes if VF/VT persists. Follow with an infusion of 1 mg/min for 6 hours, then 0,5 mg/min for 18 hours if VF/VT reverts (maximum total dose <2,2g / 24 hours).

8. **SINUS BRADYCARDIA**



Recognition:

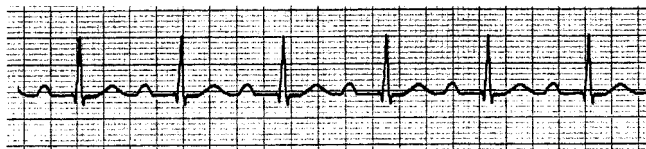
Rate < 60/minute. P waves identifiable before each QRS complex.

Management:

1. Search for and treat underlying causes of bradycardia, such as hypoxia, hypothermia, head injuries (Cushing reflex), hyperkalaemia, hydrogen ion (acidosis), hypotension, heart transplant or just normal healthy athletes
2. Provide supplemental oxygen, place patient on monitoring (ECG, BP, and pulse oximetry), and obtain IV access.
3. If the patient is unstable:
 - **Atropine:** 0,5mg IV every 3-5 minutes if necessary (Maximum - 3mg). Do **not** give atropine to patients with hypoxia, hypothermia, head injury or heart transplants.

- **Transcutaneous pacing** (if available): Particularly useful for 2nd degree (Mobitz Type 2) and 3rd degree heart blocks. Arrange for transfer to a cardiac centre for transvenous pacing as appropriate.
- **Adrenaline:** Can be used as an alternative to pacing, or if pacing ineffective. Start with 2 - 10ug/min and titrate according to response.
- **Dopamine:** can be used as an alternative to adrenaline. Start with 2 - 10ug/kg/min and titrate according to response.
- **Glucagon:** May be useful for severe bradycardia caused by beta blocker or calcium channel blocker overdose. Give 3mg initially, followed by 3mg/hour titration.
- **High dose Insulin:** Useful for beta blocker and calcium channel blocker overdose. Supplement glucose and potassium.
- **Theophylline:** May be of value for bradycardia caused by inferior myocardial infarction, spinal cord injury or cardiac transplants. Administer 100 - 200mg IV slowly over 30 minutes.

9. FIRST DEGREE HEART BLOCK



Recognition:

P wave before each QRS, but PR interval is consistently prolonged ($> 0.2\text{sec}$)

Management:

- First degree heart block can be a normal physiological variant, but can be caused by a variety of conditions such as vasovagal reactions, rheumatic fever and drugs such as beta blockers, calcium channel blockers and digitalis.
- Life-threatening conditions associated with a 1st degree heart block include myocardial ischaemia (especially inferior and RV MI), hyperkalaemia and hypermagnesaemia.

10. SECOND DEGREE HEART BLOCK - PROXIMAL TO BUNDLE OF HIS (Mobitz Type I - Wenckebach)



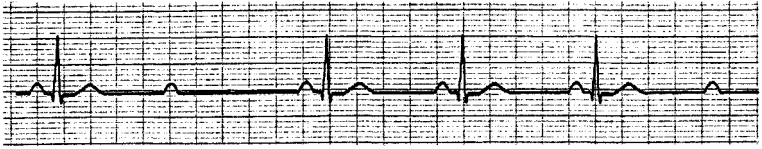
Recognition:

PR interval progressively lengthens, until a QRS is dropped (Non-conducted P wave). More P waves than QRS complexes.

Management:

- Is generally benign unless the patient is symptomatic.
- Common causes are vasovagal reactions, drugs such as beta blockers, calcium channel blockers and digitalis, and right coronary artery disease.

11. SECOND DEGREE HEART BLOCK - DISTAL TO BUNDLE OF HIS (Mobitz Type II)



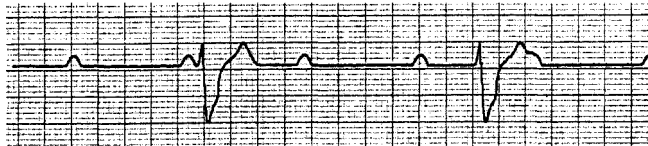
Recognition:

PR interval is normal, but there are more P waves than QRS complexes as some P waves are not followed by a QRS complex.

Management:

- Usually caused by disease of the left coronary artery.
- Pacing is usually indicated, as atropine is seldom effective.
- If pacing unavailable or ineffective, consider adrenaline or dopamine infusion.

12. THIRD DEGREE (COMPLETE) HEART BLOCK - DISTAL TO BUNDLE OF HIS (Total AV Dissociation)



Recognition:

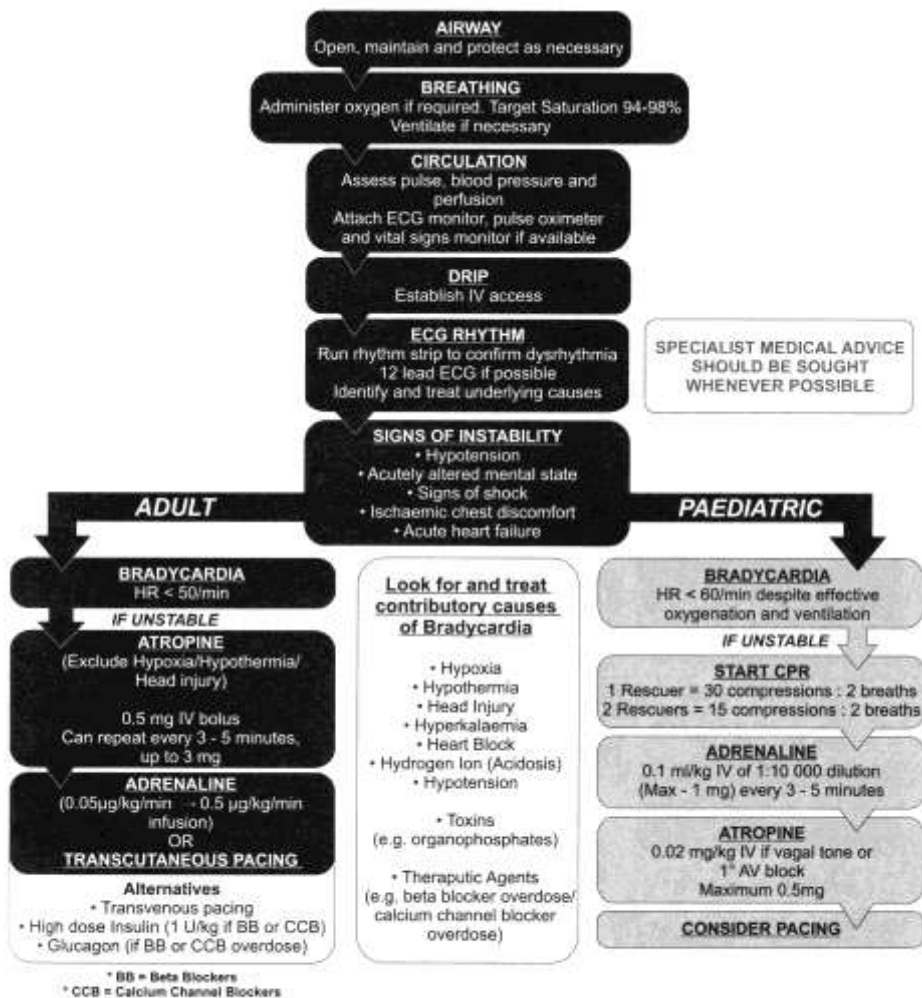
More P waves than QRS complexes. No relationship between the P waves and QRS complexes (The atrial rate and ventricular rates are different).

Management:

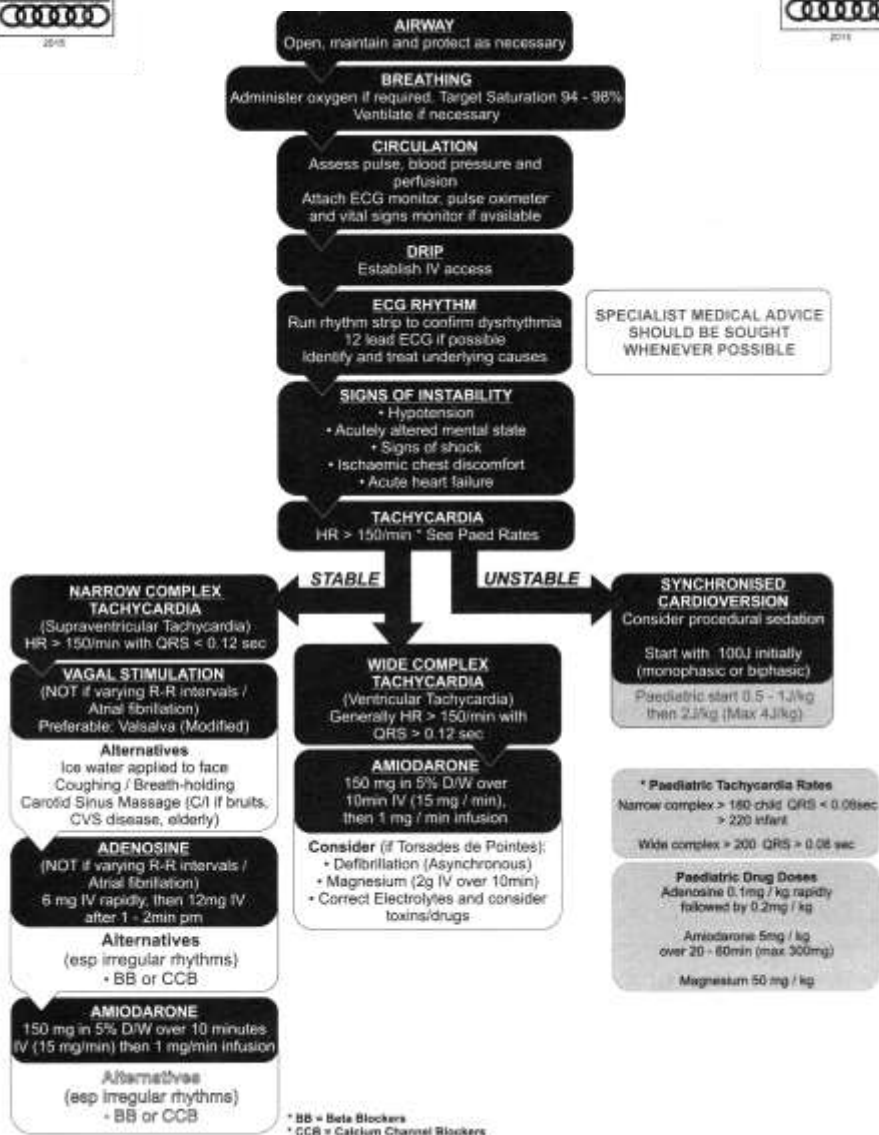
- Usually caused by disease of the left coronary artery. Occasionally caused by digitalis.
- Pacing is indicated, as atropine is ineffective unless the QRS complexes are narrow.
- If pacing unavailable or ineffective, consider adrenaline or dopamine infusion.
- Never give amiodarone or lignocaine to these patients.

The accompanying **Bradycardia Management Algorithm** and **Tachycardia Management Algorithm** provides a supplementary approach

BRADYCARDIA MANAGEMENT ALGORITHM



TACHYCARDIA MANAGEMENT ALGORITHM



SECTION 1.6 ACUTE HEART FAILURE

Definition

- Acute Heart Failure (AHF) is defined as a rapid onset or change in the signs and symptoms of heart failure

Clinical Presentation

- Patients may present as a medical emergency such as acute pulmonary oedema
- AHF may present as either new onset heart failure or more often as a worsening of pre-existing chronic heart failure
- Common features of AHF irrespective of cause are pulmonary and systemic congestion.

Causes of Acute Heart Failure

- Acute decompensation of pre-existing chronic heart failure (CHF)
- Acute coronary syndromes
- Hypertensive crisis
- Acute arrhythmias (atrial fibrillation, atrial flutter, SVT, ventricular fibrillation, ventricular tachycardia) usually in the setting of underlying heart disease
- Acute valvular regurgitation (infective endocarditis)
- Valvular disease (aortic valve stenosis and mitral stenosis)
- Acute myocarditis
- Cardiomyopathies
- High output states (sepsis with organ dysfunction, thyrotoxicosis, anaemia)

Précipitants of Acute Heart Failure

- Lack of adherence to medical therapy in CHF
- Volume overload (often precipitated by excessive fluid ingestion)
- Infections
- Renal dysfunction
- Anaemia
- Arrhythmia (eg. aortic stenosis, mitral stenosis)

Symptoms and Signs of Congestion

- Dyspnoea; orthopnoea or paroxysmal nocturnal dyspnoea
- Jugular venous distention
- Peripheral oedema
- Pulmonary crackles
- Positive abdominal-jugular reflex
- The most important sign is an S3 gallop - usually a summation gallop)

Goals of Treatment of Acute Heart Failure

- Alleviate symptoms
- Improve low oxygenation
- Improve haemodynamic parameters

Summary of Therapeutic Approaches in Acute Heart Failure

Acute Heart Failure where Systolic Blood Pressure (SBP) is > 140 mmHg

It is important to assess whether the patient is fluid overloaded: If acute pulmonary congestion in the setting of a hypertensive crisis, acute myocardial infarct, severe diastolic dysfunction or negative

pressure pulmonary oedema the patient is generally not fluid overloaded preceding the event as such they are usually fluid depleted

- If low oxygen saturation (due to pulmonary oedema): Give oxygen. Strongly consider positive pressure ventilation (non- invasive or mechanical ventilation if still hypoxaemic)
- For pulmonary congestion in the setting of fluid overload- ie the presence of pedal oedema and an elevated JVP: Use low doses of IV diuretics preferably by infusion of furosemide 1- 5mg/hour or if not possible (boluses of 20-40 mg). If secondary to severe hypertension administer labetalol and oral antihypertensives (see section on hypertensive crisis). If in the setting of myocardial ischaemia or with fluid overload consider nitroglycerine 10 - 20ug/min to reduce preload and to improve coronary perfusion, increasing dose to 200 ug/min, or isosorbide dinitrate 1 mg/hr, up to 10 mg/hr)
- Restoration of sinus rhythm or achievement of satisfactory heart rate control in the presence of atrial fibrillation or other tachyarrhythmias

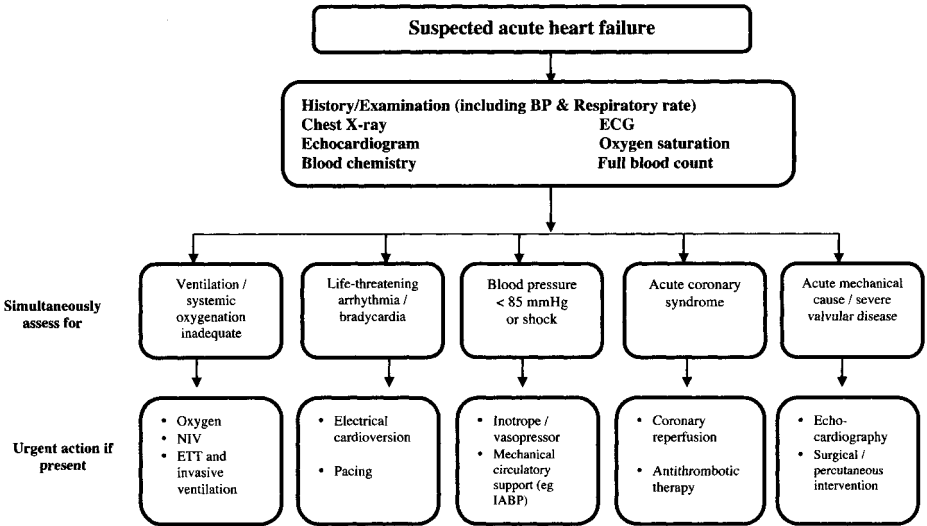
Acute Heart Failure with Normal SBP (90 - 140 mmHg)

- Improvement of fluid overload using IV diuretics preferably by infusion of furosemide 1-5mg / hour)
- Reduction of elevated cardiac filling pressures and afterload (if SBP > 100 mmHg) using low doses of ACE inhibitors with close monitoring of BP
- Improvement of peripheral organ perfusion using IV inotropes (eg dobutamine) in cases where there are features of low output and poor peripheral organ perfusion. A beta blocker such as carvedilol can be cautiously introduced when the pulmonary oedema is resolved ie 3.125mg daily and gradually increased

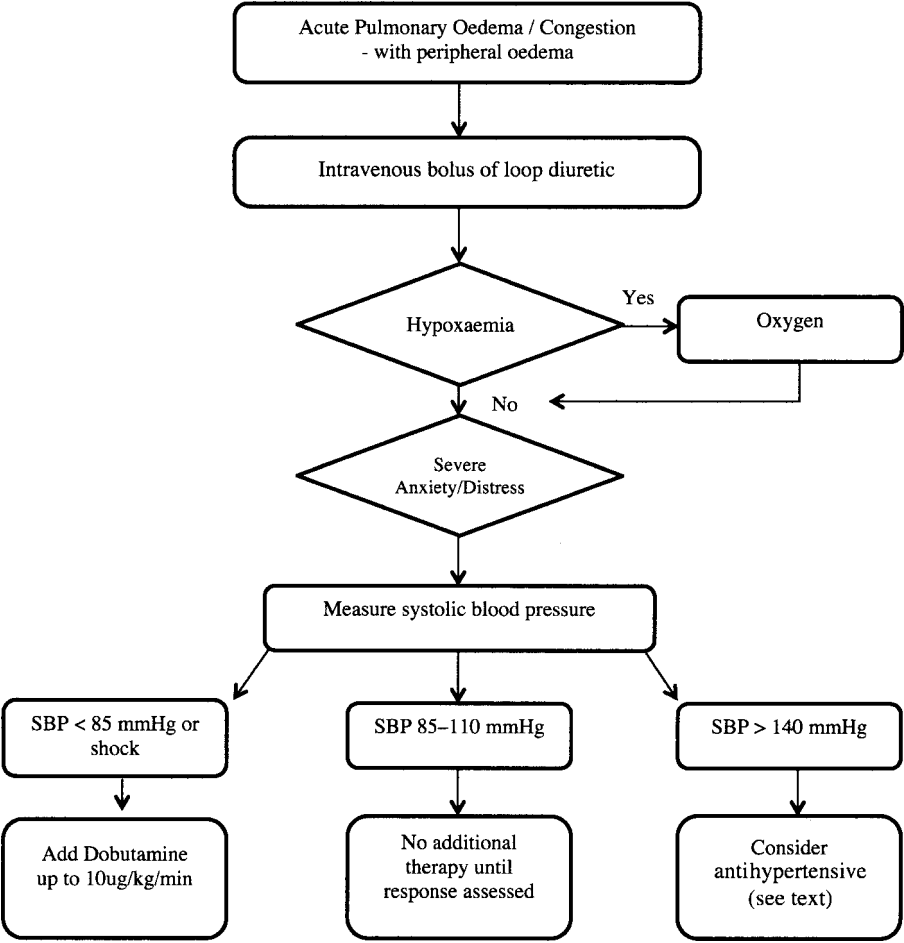
Hypotensive Acute Heart Failure (SBP < 90 mmHg)

- Restoration of SBP and peripheral hypoperfusion using inotropes (dobutamine) and/or vasopressors (adrenaline). The latter is seldom required unless there is acute right heart failure as in an RV infarct
- Non- invasive or mechanical ventilation if worsening and not responding to IV diuretics
- Consider dialysis for improvement of oligo- or anuric patients who are overloaded and not responding to conventional treatment. Remember that if oedematous and hypotensive, gentle diuresis allowing time for redistribution of interstitial fluid to the intravascular compartment actually improves renal function and urine output (by decreasing intra-abdominal pressure) and blood pressure
- When the cause of hypotensive AHF is ACS - urgent reperfusion is the first line therapeutic option

Initial assessment of patient with suspected acute heart failure



Algorithm for management of acute pulmonary oedema/congestion



Ref: European Heart Journal (2012) 33,1787-1874
European Journal of Heart Failure (2012) 14,863-869

SECTION 1.7 SHOCK

CLINICAL SIGNS OF THE THREE SHOCK STATES

Type of Shock	Distributive (Early stages)	Hypovolaemic	Cardiogenic
PROFILE (** = predominant feature)			
Blood Pressure	Decreased	Decreased	Decreased
Flow (cardiac output)	Increased	Decreased	** Decreased
Afterload (Systemic Vascular Resistance)	** Decreased	Increased	Decreased
Preload	Decreased	**Decreased	Increased or Decreased

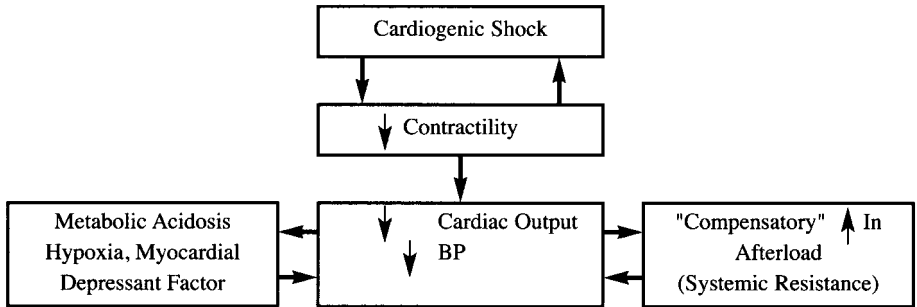
CLINICAL OBSERVATIONS

Peripheral circulation	Vasodilated	Vasoconstricted	Vasoconstricted
Peripheral cyanosis	Not usual	Probable	Probable
Pulse	Bounding	Weak, thready	Weak, thready
Central venous pressure	Not raised	Not raised	Usually raised
Cardiac auscultation	None, except from pre-existing cardiac pathology	None, except from pre-existing cardiac pathology	S ₃ Gallop, murmur, rub

LABORATORY INVESTIGATIONS

ECG	Normal	Normal unless traumatic shock complicated by myocardial injury	Abnormal
Chest x-ray	Oedema possible	Clear	Usually oedema, heart size
Central venous oxygen content			
Arterial oxygen			
Cardiac output			

SECTION 1.8 CARIOGENIC SHOCK



DIAGNOSIS

- (1) Hypotension (systolic <80 mmHg, mean <50 mmHg) and
- (2) Evidence of low perfusion:
 - (a) decreased cerebral flow - obtunded, restlessness, confused
 - (b) decreased renal flow - oliguria
 - (c) decreased peripheral flow - cold, clammy skin
- (3) Often associated right ventricular failure or pulmonary oedema.

CAUSES OF CARDIOGENIC SHOCK

- (1) Myocardial Infarction
 - (a) Loss of >50% of myocardium (massive or cumulative repeated infarctions).
 - (b) Mechanical causes (VSD, papillary muscle rupture, rupture of ventricle with tamponade).
 - (c) Primary tachy- or bradyarrhythmias.
 - (d) Right ventricular infarction.
- (2) Terminal Event in End-Stage Cardiac Failure - of any cause:

Precipitating factors:

Ischaemia
Thyrotoxicosis
Arrhythmias
Pulmonary embolism
Infections
Drug toxicity (Ca⁺⁺ antagonists, B blocker, digitalis intoxication)
Anaemia

- (3) Valvular Disease
 - (a) Mitral regurgitation
 - (b) Aortic regurgitation
 - (c) Aortic stenosis

CARDIOGENIC PULMONARY OEDEMA

Cardiac/Fluid overload:

1. Cardiac failure
2. Fluid overload (manifested by peripheral oedema)

Common Presentation:

Dyspnoea/Tachypnoea/Orthopnoea

Hypoxia (reduced saturation)

Productive cough with pink, frothy sputum Wheezing, x-

ray signs of pulmonary oedema **Cardiac (Diastolic or**

Systolic Failure)

Diastolic failure (backward failure) presents with pulmonary oedema. Often has a normal ejection fraction- particularly in females

Systolic failure (forward failure) presents with hypoperfusion, decreased urine output, lethargy, confusion

1. May be hypertensive or shocked
2. If biventricular failure: raised JVP, oedema, hepatomegaly, as well as gallop rhythm (S3), crackles in chest
3. Evidence of cardiac disease: hypertension, valvular disease, CMO, ischaemia or congenital (VSD, PDA, coarctation)

Initial Management

1. Maintain airway
2. Fowler's position (Unless hypotensive or comatose)
3. Administer oxygen to keep $pO_2 > 60$ mmHg (8 Kpa)
4. Correct acid-base and electrolyte disorders
5. Diagnose and correct arrhythmias (cardiology CONSULT)

Cardiac Failure

1. CPAP (5-10cm H₂O) or non-invasive ventilation (Respiratory CONSULT)
2. Furosemide if oedematous - initial dose of 20 - 40 mg IV, may be repeated in 10 - 15 minutes if symptoms persist, or infuse.
3. Inotropic support if hypotensive or decreased urineoutput) Dobutamine 5-10 μ g/kg/min)
4. If hypertensive, reduce BP with labetalol and add oral antihypertensives.
5. IV TNT or sublingual nitrate particularly if ischaemia
6. Introduction of afterload reducers: ACE inhibitors ((Perindopril 2-4 mg daily; enalapril 2.5 mg BD)
7. Cautious introduction of alpha/beta blocker (Carvedilol 3.125 mg daily, then BD): Cardiology CONSULT
8. Treat the underlying cause

POOR RESPONSE: (Consult Cardiac ICU)

1. Intubate and ventilate with PEEP
2. Dialysis if in renal failure

SECTION 1.9 ACUTE ANAPHYLAXIS

Anaphylaxis is an acute, severe, life-threatening multisystem hypersensitivity reaction. Manifestations of anaphylaxis may occur within seconds or minutes after exposure to a causative antigen.

Any antigen capable of activating IgE can trigger anaphylaxis, but non-immunological (anaphylactoid) reactions can present identically and are treated the same way. Severe reactions can occur without any documented prior exposure.

Common causes of anaphylaxis include antibiotics, aspirin, non-steroidal anti-inflammatories, contrast media, latex, insect stings, nuts, seafood, wheat and many other substances.

The reaction may become life-threatening with the development of any 2 or more of the following manifestations:

- a) Acute respiratory difficulty
- b) Signs of shock/hypotension
- c) Involvement of skin/mucosal tissue
- d) Gastro-intestinal symptoms
- a) **Acute Respiratory Difficulty**
 - Progressive swelling of face, neck, soft tissues
 - Hoarseness, stridor, wheezing, dyspnoea, distress
 - Reduced peak expiratory flow
 - Hypoxaemia, cyanosis.
- b) **Signs of Shock/Hypotension**
 - Lightheadedness, pale, clammy, hypotonia, syncope
 - Systolic BP < 90mmHg, or > 30% decrease from patient's baseline
 - Incontinence
- c) **Involvement of Skin/Mucosal Tissue**
 - Generalized pruritis, urticaria
 - Pale or flushed
 - Swollen face, lips, tongue, uvula
 - Rhinitis (often an early sign)
- d) **Gastro-intestinal Symptoms**
 - Crampy abdominal pain
 - Nausea, vomiting, diarrhoea

Reactions may be slow, progressive, or rapidly fatal within minutes. Manifestations may be delayed, or persist for more than 24 hours, or may recur (biphasic) up to 36 hours after initial onset. Generally, the shorter the interval between exposure and reaction, the more severe the reaction.

MANAGEMENT OF ANAPHYLAXIS

1. **Remove** or stop the precipitating agent immediately.
2. **Intramuscular Adrenaline:**
 - Inject 0,3 - 0,5ml of adrenaline (1:1000) intramuscularly into the anterolateral aspect of the thigh if the patient develops progressive systemic features. (Do **NOT** inject subcutaneously, as the absorption may be delayed).
 - Repeat every 5-15 minutes if no improvement (Between 18 and 35% of patients with anaphylaxis may require a second dose of adrenaline)
3. **Oxygen:**
 - Provide high flow oxygen via rebreather mask as soon as possible.
4. **Maintain a Patent Airway:**
 - Position patient semi-Fowler's (unless hypotensive) to assist breathing. Monitor respiratory parameters and vital signs (BP, Sats and ECG) continuously.
 - Beware that progressive airway obstruction may occur, necessitating emergency intubation or even cricothyrotomy.
5. **Establish Rapid IV Access:**
 - Rapidly infuse 1-2 litres of crystalloid (Ringers or other balanced salt solution) if hypotensive or unresponsive to IM adrenaline.
 - Repeat the IV infusion if necessary, as large amounts may be required
 - Aim to keep the SBP > 90mmHg.
6. **Intravenous Adrenaline:**
 - IV adrenaline is potentially hazardous in anaphylaxis, and should only be considered if life-threatening hypotension persists despite the administration of intramuscular adrenaline and aggressive fluid resuscitation.
 - Dilute 1mg adrenaline in 200ml normal saline, and slowly infuse at 1 ml/minute (5ug/min) with continuous ECG monitoring.
 - Titrate the adrenaline infusion (0,1 - 1ug/kg/min) in accordance with the patient's response.
7. **Antihistamine:**
 - Administer an antihistamine such as promethazine 25mg IM or slowly IV.
8. **Nebulized Bronchodilators:**
 - Nebulized salbutamol (5mg) together with ipratropium (0,5mg) should be given every 15-20 minutes if bronchospasm is a major feature, and especially if the patient is on beta blockers.
9. **Corticosteroids:**
 - Particularly useful for preventing or shortening protracted reactions.
 - Administer hydrocortisone 200mg IM or slowly IV.
10. **Glucagon:**
 - Administer 1-2mg IM or slowly IV every 5 minutes if the patient is unresponsive to adrenaline and fluids, and especially if the patient is on beta blockers.
 - Watch out for vomiting and hyperglycaemia.

11. **H2 Receptor Antagonist:**
 - Consider the administration of either ranitidine (50mg IM or slowly IV, diluted in 20ml over 2 minutes) or cimetidine (300mg IM or slowly IV, diluted in 20ml over 2 minutes).
12. **Admit for Observation:**
 - Recurrences may occur, and therefore the patient should be admitted and observed for 8-24 hours.
 - Antihistamines and oral steroids should be continued for the next 3 days after discharge.
13. **Prevent Recurrence:**
 - Identification of the cause to prevent recurrence is important, and arrange for a “Medic Alert” bracelet for the patient.
 - It is essential to prescribe and educate the patient and relevant family members on the use of a self-injectable adrenaline device (EpiPen). All patients having had a severe reaction should have such a kit on their person at all times.

Preventative Measures 1. Ask about Allergies

The most important aspect in the management of anaphylaxis is to try and prevent the condition from re-occurring. Obtain details of any previous suspected allergic or adverse reactions. If there is any doubt, carefully weigh up the absolute necessity of administering the medication, especially parenterally, and consider other alternatives rather. It is always wise to keep a written record of the questions asked and the patient’s reply.

2. **Advanced Preparation**
Before administering medications, ensure that you have the necessary skill and emergency equipment to handle an adverse reaction. Is a properly stocked advanced life support kit or trolley readily available, and are the staff adequately trained to use the equipment? An acute anaphylactic reaction can develop literally within seconds, and is one of the most frightening of medical emergencies for both patient and doctor. The resuscitation skills of those present may well be put to the ultimate test.
3. **Anticipation**
It is worthwhile asking patients to wait in a dedicated area for up to 30 minutes, especially following parenteral administration of medications. Inform patients of the early symptoms of anaphylaxis, and to report these immediately if they occur. Stop further administration at the first signs of anaphylaxis. Severe reactions may occur (although rarely) even up to 24 hours after exposure.
4. **Accessibility**
In the management of the patient, it is essential to ensure the safety of the rescuer as well as the patient. Following a bee sting attack, for example, make sure that the scene is safe first. Do not squeeze the venom sac when removing the stinger; use a credit card or blunt end of a knife instead of your fingers.

5. **Accurate Assessment**

The symptoms and signs of an anaphylactic reaction must be differentiated from the non life-threatening features of a simple allergic reaction (urticaria, pruritus, conjunctivitis, rhinitis, etc) and from other conditions such as vasovagal attacks (syncope), hypoglycaemia, panic attacks, etc. Recognition of the development of the above-mentioned manifestations of anaphylaxis is of utmost importance, since deaths have been reported within hours and even minutes of the onset of symptoms. Confirmation of the anaphylactic reaction can be obtained by measuring mast cell tryptase levels after resuscitation has commenced, and again 1-2 hours after the onset of symptoms (most important sample), and again after 24 hours (for baseline levels).

6. **Ambulance**

In the pre-hospital setting, the rapid deployment of advanced life support paramedics will be required in order to provide the necessary equipment and expertise, in view of the potential rapid progression of this condition.

Knowledge of the telephone numbers of an appropriate ambulance service is fundamental, yet surprisingly few people have these numbers readily available. Emergency numbers should be clearly visible and securely taped onto all telephones, as there may not be sufficient time to start searching for a telephone directory or a misplaced piece of paper.

Family members should be taught CPR. Oxygen therapy, airway protection and other life support procedures can be provided by ambulance personnel according to their qualifications. It is wise for healthcare professionals to have an ampoule of adrenaline, needle and syringe safely stored in their home at all times.

Any healthcare worker involved in the administration of medications or the care of patients - whether doctor, dentist, nurse, paramedic or allied health professional - is ethically obliged to be able to appropriately manage the potential consequences of his actions - ***the anaphylactic reaction!***

The accompanying **Emergency Management of ANAPHYLAXIS** Algorithm provides a complementary approach.

Emergency Management of ANAPHYLAXIS

(Adult and Child)

ACUTE RESPIRATORY DIFFICULTY
(Piloerecive Swelling, Stridor, Wheezing, Coughing
and/or
SIGNS OF SHOCK/HYPOTENSION
(especially if skin or mucosal changes are present)

Early (first line) treatment

ADRENALINE
1mg/ml (1:1000) - 0.01mg/kg IM (Maximum - 0.5ml IM)
or
Auto-Injector
> 6 yrs - 0.3 ml IM
< 6 yrs - 0.15 ml IM
REPEAT EVERY 5 - 15 MINUTES IF NO IMPROVEMENT

OXYGEN - MONITORS - IV ACCESS

- High flow oxygen
- Maintain patent airway (intubate/Cricothyotomy if necessary)
- BP, SpO₂, ECG monitoring
- Lie patient supine with legs elevated if hypotensive
- High flow IV line

Adjunctive treatment

H1 ANTIHISTAMINE

Promethazine
> 12 yrs - 25 mg IM or slow IV
8 - 12 yrs - 12.5 mg IM or slow IV
2 - 8 yrs - 6.25 mg IM or slow IV
(Avoid if < 2 yrs old and low BP)

CRYSTALLOID
(e.g. Ringers/Lactol)

Rapid infusion of 1 - 2 litres (20 ml/kg for children)
if no response to adrenaline
Repeat IV infusion as necessary, as large amounts
may be required
Adrenaline infusion (0.1 - 1 ug/kg/min) ONLY if
unresponsive to IM adrenaline and fluids

NEBULISED BRONCHODILATORS

(if severe bronchospasm, and especially if
on beta blockers)

Salbutamol

> 6 yrs - 5 mg every 15 - 20 mins
< 6 yrs - 2.5 mg every 15 - 20 mins
WITH Ipratropium
> 6 yrs - 0.5 mg every 15 - 20 mins
< 6 yrs - 0.25 mg every 15 - 20 mins

H2 RECEPTOR ANTAGONIST

Ranitidine
Adult - 50 mg IM or slow IV (diluted in 20 ml
over 2 min)
Child - 1 mg/kg (Max - 50 mg)
OR
Cimetidine
Adult - 300 mg IM or slow IV (diluted in 20 ml
over 2 min)
Child - 5 mg/kg (Max - 300 mg)

GLUCAGON

Adult - 1 - 2 mg IM or slow IV every 5 mins if
unresponsive to adrenaline, and especially
if on beta blockers
Child - 20 ug/kg (Max - 1 mg)
(Look out for vomiting and hyperglycaemia)

CORTICOSTEROIDS

Hydrocortisone
> 12 yrs - 200 mg IM or slow IV
8 - 12 yrs - 100 mg IM or slow IV
1 - 8 yrs - 50 mg IM or slow IV
< 1 yr - 25 mg IM or slow IV

Recommender Information

Q When is it appropriate to initiate treatment for Anaphylaxis?

A All patients with severe allergic reactions should be treated. Do not wait for symptoms to progress.

Q What should happen once a patient is resuscitated?

A Patients should be taken to hospital for further treatment.

Q What is the discharge protocol for patients leaving the hospital?

A (Refer patients home with)
1. A written 0-3 discharge
2. An Anaphylaxis Emergency Action Plan
3. An adrenaline auto-injector

Q Why are Antihistamines considered adjunctive treatment?

A H1 antihistamines may relieve itching and allergic skin reactions or relieve the bronchospasm of anaphylaxis. However, antihistamines alone are not enough. Immediate resuscitation for airway, breathing and circulation is essential. Antihistamines are not a substitute for adrenaline and should only be given if symptoms are severe and the patient is not in a life-threatening situation.

SECTION 1.10 SEPSIS AND SIRS (SYSTEMIC INFLAMMATORY RESPONSE SYNDROME)

SEPSIS (confirmed / suspected infection) or SIRS:

Pyrexia (Temp $>38^{\circ}\text{C}$ or $<36^{\circ}\text{C}$)

WCC <4 or >12

Tachypnoea (RR >20) or decreased pCO_2

Tachycardia (HR >90).

- New sepsis definitions: ward patient If SIRS criteria present assess for presence of sepsis -then do a **qSOFA** to prognosticate: Respiratory Rate $> 22/\text{min}$
- Altered mentation
- Systolic BP < 100

If > 2 patient should be transferred to ICU for further management **Septic**

Shock:

Clinical construct of sepsis with:

- *Persistent hypotension requiring vasopressors to maintain MAP $>65\text{mmHg}$*
- *Lactate $>2\text{ mmol/L}$*

Organ Dysfunction may ensue:

Circulatory - hypotension, reduced CO, vasodilatation

Respiratory - Acute Respiratory Distress Syndrome (ARDS)/Acute Lung Injury (ALI), respiratory failure.

Renal - Oliguria, acute renal failure Haematologic - coagulopathy Neurological - encephalopathy Hepatic dysfunction

MANAGEMENT

1. Antibiotics: adequate and as early as possible
2. Ensure adequate oxygen delivery: Fluid (Ringers' Lactate); - but beware of overloading patient. Do not continue crystalloids in the presence of sacral oedema inotropes, vasopressors if unresponsive to fluid (usually adrenaline initially), oxygenation/possibly mechanical ventilation
3. Steroids: (Hydrocortisone 100 mg 8 hrly) if catecholamine-dependent despite fluid, until inotropes weaned
4. Insulin: Keep glucose $<10\text{ mmol/L}$.
5. Ventilate according to ARDS-NET criteria (See below)
6. Early enteral nutrition
7. Source control: Identify and eradicate focus of sepsis
 - a) Monitor BP, Pulse, and urine output hourly.
 - b) Monitor arterial blood gases and electrolytes.

ARDS may be Primary or Secondary

Primary: pneumonia, fat embolism, aspiration, amniotic fluid embolus, near drowning, contusion, heroin (idiosyncratic), inhaled toxins (e.g. chlorine gas), paraquat poisoning.

Secondary: severe sepsis, pancreatitis, head injury, shock, major trauma, blood transfusion post cardiopulmonary bypass),